FORM AND RISK CLASS DISTRIBUTION AND ITS EFFECT ON ESTIMATED PRODUCT RATIOS IN COMMERCIAL SPECIES COMMONLY FOUND IN NORTHEASTERN FORESTS AND A COMPARATIVE ANALYSIS OF CONTRASTING FIELD INVENTORY METHODOLOGIES

By

Ryan Smith

B.A. International Development and Social Change, Clark University, 2009

M.A. Community Development and Planning, Clark University, 2012

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Advisory Committee:

Aaron Weiskittel, Professor of Forest Biometrics and Modeling, Advisor

Ian Prior, Senior Inventory and Planning Manager, Seven Islands Land Company

Laura Kenefic, Research Forester, U.S. Forest Service, Northern Research Station

Anil Kizhakkepurakkal, Associate Professor of Forest Operations

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FORM AND RISK CLASS DISTRIBUTION AND ITS EFFECT ON ESTIMATED PRODUCT RATIOS IN COMMERCIAL SPECIES COMMONLY FOUND IN NORTHEASTERN HARDWOOD FORESTS AND A COMPARATIVE ANALYSIS OF CONTRASTING FIELD INVENTORY METHODOLOGIES

By Ryan Smith

Thesis Advisor: Dr. Aaron Weiskittel

An Abstract of the Thesis Presented
in Partial Fulfillment of the Requirements for the
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Estimating product ratios in hardwood forests is difficult due to the highly complex stem forms and diverse growth and regeneration strategies found in hardwood species. This is of particular concern for landowners with large ownerships requiring robust product potential estimates as part of both short-term tactical harvest planning and long-term growth and yield projections. Hardwood-dominated forests are generally managed for high-value sawlog and veneer products. This makes it increasingly important to identify strategies for improving hardwood management to meet economic (maximizing sawlog recovery) and ecological (maintaining healthy and diverse forests) goals.

This study seeks to achieve two primary goals. The first is to further develop predictive models for sawlog potential utilizing a stem form and risk framework that can be easily added to existing field inventory protocols. The second is to compare commonly used field merchandising techniques and a commonly used regional growth model with the form and risk approach to understand how each method affects plot-level predictions of sawlog volume, stand value, and the variability of sawlog volume compared to merchantable volume in forest inventories.

Results from this study show that sawlog potential in hardwood forests can be effectively modeled using easily collected data on stem form and risk in a way that both improves short-term estimates of available sawlog volume and long-term projections of sawlog potential in future stand conditions. In past studies, stem form and risk have been shown to be important predictors of diameter growth and mortality in growth and yield modeling. This study further demonstrates the importance of this data by showing that they can be effective predictors of sawlog presence and volume in commercially important hardwood and softwood species.

Robust models predicting sawlog presence and volume in 14 commercial species were developed in this study. The results from these model predictions using stem form and risk were compared to estimates of sawlog volume using four common field inventory methods for predicting product ratios and predictions from a commonly used regional growth model, the Forest Vegetation Simulator (FVS), in ten northern hardwood stands. Statistically significant differences were found among methods, and all field methods demonstrated reasonable estimates of available sawlog volume and stand value while FVS did not. We also found that there was significantly more variability among sample plots when measuring sawlog volume than when measuring merchantable volume. This showed the importance of utilizing an efficient field methodology that can be applied with the intensity required to lower error rates in forest inventories.

Form and risk observations are a potentially effective alternative to commonly used field methodologies. They provide reasonable estimates of sawlog potential, an easy framework to integrate into existing inventories, and a tool that may be used to predict sawlog potential in future stand conditions, thus improving the performance of existing forest growth and yield models and capturing an accurate picture of current and future hardwood forest conditions for both short-term tactical and long-term harvest planning.

DEDICATION

This work is dedicated to my wife Randi for her endless support and encouragement and to my late grandfather, Willard Stearns Caswell Jr., who instilled in me an appreciation and love for Maine's forests.

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PROLOGUE

It is an uncertain time concerning the management of Maine's large and diverse range of hardwood-dominant and mixed-wood forests. Two competing forces are pulling on management objectives and options. The first of these forces has been a challenging one. There has been a hard and relatively abrupt shift in the timber industry that has led to a collapse of the paper industry and the pulp markets it provided (Crandall et al., 2017; Listo, 2018). While this change has been spread out over the course of three decades, it is abrupt relative to the long timescale of forest growth and regeneration of northern hardwoods.(Boucher et al., 2006). The effects of these changes are being felt all over a region that for the last century has been managing forests largely to feed these pulp markets. Currently, particularly for hardwood species, saw timber and veneer markets are increasingly the economic drivers and there is little support from pulp markets that once provided an economic relief valve for pulp products that make up a majority of merchantable stand volume and harvests (*The Economic Importance Of Maine's Forest-Based Economy 2013*, 2013).

The second competing force is one that has the potential to dramatically improve economic outcomes for landowners in these new market conditions. This force is the increased access to new technology, and in particular the combination of LiDAR-driven advanced forest inventories with improved regional growth and yield models (Ayrey et al., 2021; Babcock et al., 2016). These tools provide landowners with significant new advantages as they pivot away from the old economy and towards new management practices that confront the new economic realities.

Despite this innovative technology, a key problem remains. While these emerging technologies offer landowners new power to understand and manage their forests, they don't currently have the capacity to answer two fundamental questions: firstly, how much sawtimber is currently standing in Maine's

hardwood forests, and secondly, how do we meaningfully target our resources to measure, manage, and harvest it in a way that meets landowners' economic and ecological goals?

This problem is compounded by the fact that forests in Maine are increasingly dominated by hardwoods with trends pointing towards decreased natural regeneration of red spruce and balsam fir (Morin et al., 2007). Softwoods like spruce and fir have more consistent stem forms and decreased variability when compared to hardwoods, which makes their product potential easier to predict. The complex mix of growth and regeneration strategies seen in hardwood species makes predicting stem form and product potential a difficult task, as demonstrated by Mark Castle in previous work that has looked at this issue (Castle et al., 2017).

While enhanced forest inventories utilizing LiDAR and improved forest growth models can provide landowners with better estimates of total merchantable volume and forest type, the potentially poor correlation between merchantable volume and sawlog volume due to the complexity of hardwood stem form renders this data insufficient for landowners who seek a way to achieve both their economic and ecological goals. Robust estimates of current and future sawlog volume are necessary to both efficiently plan the time of harvests and to apply appropriate silvicultural treatments in hardwood forests. This means that improved systems for measuring this sawlog potential need to be developed and integrated into forest inventory systems, and this integration needs to be done in a way that doesn't unnecessarily burden the resources available for forest measurement.

Though new technologies allow landowners to design inventories more effectively by decreasing sampling error and increasing their broad understanding of the land base, there is still a need for boots on the ground. Large landowners rely on field inventory data for developing both short-term tactical management and harvesting plans and long-term management plans that focus on forest growth and future yield and the planning of treatment regimes to maximize future value.

Landowners have confronted the challenge of collecting field inventory data that accurately represents their land base largely by utilizing field inventory methods that were in widespread use prior to the collapse of the pulp market. These methods provide static estimates of sawlog potential, and their variability and error rates have not been examined in detail. It is possible that an effective alternative can be developed that is applicable to merchantable species in the region. Maine forests are unique in the number of merchantable species that are managed for, along with the ability of these species to regenerate naturally post-harvest. The forest types and silviculture in the region make it more difficult to predict product ratios than in a plantation setting, especially in forests dominated by hardwoods.

This study seeks to achieve two main objectives. The first is to further develop predictive models for sawlog potential utilizing proposed measurement techniques that can be easily added to field inventories. The second is to compare existing field merchandizing methodologies and a commonly used regional growth model, FVS, with the models developed in the first objective to understand how each method affects predictions of sawlog volume, stand value, and the variability of sawlog volume compared to merchantable volume. Understanding the variability in sawlog volume allows for inventory managers to design inventories to decrease error by increasing sampling intensity and allocating resources to areas where high variability requires more intensive measurement.

The hope is that the knowledge gained from these objectives will help land managers integrate field merchandizing methods into their inventories in a way that not only allows for a realistic examination of standing sawlog potential, but also allows for growth and yield projections to integrate this data into predictions of future conditions to improve long-term management outcomes. Given the poor correlation between total merchantable volume and standing sawlog volume, knowing standing sawlog volume may help land managers know when stands are operational and which management practices can increase sawlog volume as a proportion of total merchantable volume.

CHAPTER 1

THE ROLE OF STEM FORM AND RISK FOR PREDICTING SAWLOG PRESENCE AND PRODUCT POTENTIAL IN COMMERCIALLY IMPORTANT SPECIES IN NORTHERN HARDWOOD AND MIXEDWOOD FORESTS

Abstract

Stem form is a highly variable and complex phenomenon in hardwood and some softwood species. Stem form varies widely among species and individual trees based on regeneration strategies, competition, micro-site conditions, and possibly genetics. Existing research has examined the distribution of stem form for four commercially important hardwood species in Maine. This work found stem form to be a difficult phenomenon to predict, but that it was significant in predicting sawlog volume, diameter growth, and mortality. This analysis seeks to expand this examination of stem form to a larger collection of species using a guide developed by the Northern Hardwood Research Institute (NHRI) that looks at both stem form and risk using eight form categories and four risk categories. Form is defined as shape of the stem and risk by damage to the stem, vigor, stem form, and competition. This analysis used a large dataset of 18,003 trees from 1,794 plot locations across a broad range of conditions in the state of Maine. We found that all observed hardwood species as well as white pine, northern white-cedar, and eastern hemlock displayed high levels of variability in stem form, especially when compared to spruce and fir species, which exhibited little variation. This analysis then used climate, soil, site, tree-level, one-sided competition, and two-sided competition metrics to develop equations that predict stem form and risk, the likelihood of a stem producing a sawlog, and the proportion of sawlog volume relative to total merchantable volume in a tree. We identified redundancies between the form and risk categories defined by the NHRI and created a simplified framework with two stem forms and three risk categories. Models were created for spruce-fir, hardwoods, and other softwood species. We found that stem form and risk were hard to predict with

poor model performance for non-spruce-fir species tested. We then found stem form and risk to play a significant role in the prediction of sawlog presence in, and built models with excellent predictive power for, all hardwood species tested ($p < .001$). The model for hardwoods accurately predicted sawlog presence in the sample data 86% of the time. The spruce and fir softwood presence model had a 90% accuracy rate and the model for other softwood species had an 84% accuracy rate. Finally, we found stem form and risk to be significant predictors in the volume of sawlog material relative to total merchantable volume in all non-spruce-fir species tested. We found non-linear relationships between diameter at breast height and sawlog presence and between diameter at breast height and sawlog volume with maximum sawlog potential. One-sided competition metrics, height to diameter ratios, and climate variables proved significant depending on the species ($p <.05$). While predicting stem form and risk was unsuccessful, robust models were produced for both the prediction of sawlog presence and the proportion of sawlog volume available relative to merchantable volume. We found that collecting stem form and risk in the field using two stem forms and three risk classes was valuable for modeling sawlog presence and volume. We believe that these methods provide a simple and efficient tool for measuring product potential in Maine forests.

Introduction

The northern hardwood and mixed hardwood forests of the United States comprise 8.1 million ha of land covering a diverse mix of species and stand types with a generally cool, moist climate (Eyre, 1980; Leak et al., 2014). The region's complex geological history of repeated glaciation and millions of years of erosion has resulted in a diverse range of micro-site and soil conditions (Fisher & Binkley, 2000; Loo & Ives, 2003).

The anthropological history of northern hardwood forests is as complex as their natural history. For much of the past 300 years, the northern hardwood forests have seen high-grading

for quality timber and heavy removal cuttings for firewood and industrial uses (Ashton & Kelty, 2018). In more recent history, partial harvesting techniques including single-tree selection and group selection systems with a reliance on natural regeneration have been utilized in northern hardwood stands (Ashton & Kelty, 2018; Olson & Wagner, 2010). Contrary to the region's earlier development, a move towards sustainable forest management has been recognized in order to maintain the diverse ecological functions of the forest (Cashore et al., 2004).

These diverse forests are an important economic driver in the state of Maine, with a 2016 interagency government report estimating that the forest products industry accounts for $8.5 billion in value and 33,000 jobs (*Maine Forest-Based Economy*, 2017) . Economically important hardwood species in the region include sugar maple (*Acer saccarum* Marsh.*),* red maple *(Acer rubrum* L.), yellow birch *(Betula alleghaniensis* Britt.), northern red oak (*Quercus rubra* L.), and paper birch (*Betula papyrifera* Marsh.). A common objective in the management of these species is to extract the valuable lumber and unique veneer products that they are capable of producing (Cecil-Cockwell & Caspersen, 2015). Economically important softwoods in the region include red spruce (*Picea rubens* Sarg.), balsam fir (*Abies balsamea* L.), eastern white pine (*Pinus strobus* L.), eastern hemlock (*Tsuga canadensis* L. Carr.), and northern white-cedar (*Thuja occidentalis* L.) (Eyre, 1980; Eyre & Zillgitt, 1953). It has been demonstrated that in many parts of the forest the relative importance of hardwood species, as opposed to that of conifers, has increased over time in areas where there was intensive diameter-limit and commercial clearcutting (Olson & Wagner, 2010). In addition, climate change is expected to favor certain hardwood species over softwoods, particularly spruce-fir (Janowiak et al., 2018).

This varied history and diverse species composition has led to a complex forest structure that displays a wide degree of variability both among and within sites in the region. This is further complicated by the fact that many of these species, particularly the northern hardwoods, display a variety of stem forms that can include sweep, lean, forking, and irregular branching (Castle et al., 2017; Fortin et al., 2009). These tree forms can affect product potential and vulnerability to disturbance (Castle et al., 2017; Pelletier et al., 2016b).

Recognizing the complexity and variability of the species composition and stand conditions in the region and the economic importance of a wide variety of species, this analysis seeks to assess variability in stem form and risk across hardwood species and economically significant softwoods such as eastern white pine and Northern white-cedar. Stem form and risk will be assessed across the diverse range of stand- and site-level conditions that are a result of the region's complex geology and anthropological history. This study will then look at the role of stem form and risk, along with additional climate, soil, tree, and stand level metrics, in predicting sawlog likelihood and the percentage of merchantable volume in trees available as sawlogs. Maine's hardwood forests are an important economic driver in the state, and this study seeks to improve the management of these forests by providing landowners with additional tools for predicting product potential, stand value, and future economic potential.

Assessing Economic Value

The effectiveness and quality of sustainable forest management decisions depend on the quality of forest inventory data, which is most valuable when it includes estimates of product potential and merchantability (Borders et al., 2008). Accurately calculating predicted yield values for both long-term and short-term operational planning requires reasonable

estimates of stand values and potential product ratios (Holopainen, Mäkinen, Rasinmäki, Hyytiäinen, Bayazidi, & Pietilä, 2010a). Harvest scheduling also depends on inventory data that accurately represents the merchantable volume and product potential in managed forests. Given the importance of measuring product potential for operations and planning, this analysis will evaluate the relationship between various tree- and stand-level attributes on tree merchantability. The ability to increase accuracy and decrease variability in the prediction of product potential in timber cruises can improve yield projections, management plans, and estimates of net present value (Borders et al., 2008; Eid et al., 2004).

Predicting Product Potential

The proportion of logs in a tree that can be sawn into lumber is highly variable, and often a tree has no sawlog potential. The occurrence of sawlogs is hard to predict. Both external and internal defects vary widely between individual stems. Additionally, it is often impossible to assess both the existence and the extent of internal defect and decay prior to felling (Frank et al., 2018).

The development of tree class marking guides for northern hardwoods dates back to at least the mid-1900's, specifically by the U.S. Forest Service within the lake states region (Arbogast, 1957; *Ontario Tree Marking Guide*, 2004). These tree marking guides prioritize the identification of straight dominant or codominant trees of clear length. The system accounts not only for the merchantability of the tree, but also for its relative position within the canopy, its competitive status, and attributes like high forks. The Arbogast (1957) tree class guide addresses vigor but makes minimal reference to stem form beyond ideal straight stems. It also does not account for the potential ability of trees of a variety of stem forms to produce high

quality butt logs, even when defect, irregular form, or unclear faces are present on other parts of the stem. The guide effectively classifies trees relative to each other within a stand, but it is not designed for predicting a specific tree's sawlog volume relative to its total merchantable volume (Arbogast, 1957).

Another marking guide in common use is the AGS/UGS system (*Ontario Tree Marking Guide*, 2004). This two-class system defines a tree as acceptable growing stock (AGS) or unacceptable growing stock (UGS) and makes even broader observations than the early Forest Service marking guides (Arbogast, 1957). AGS/UGS markings are limited to whether a tree is expected to contain a high quality log and demonstrates vigor (AGS) or if it is likely to contain low quality logs and demonstrates low vigor (UGS) (*Ontario Tree Marking Guide*, 2004). This system does not account for the fact that vigor and log quality may be uncorrelated.

Specific vigor-related variables such as fruiting bodies and cracks have been shown to be predictors of sawlog potential, but without including the specific defects, the vigor of a tree has not been shown to be strongly correlated with sawlog potential or economic value (Havreljuk et al., 2014). This highlights that trees of high vigor may have external defects that prevent them from producing sawlogs, while low vigor trees may produce excellent sawlogs if extracted prior to mortality. Systems have been explored for assessing a tree's vigor and likelihood of survival, but these systems are not concerned with a tree's merchantability (Hartmann et al., 2008).

Another common marking system in the field includes the ABCD system, a system that focuses on a tree's sawlog producing potential as well as its likelihood of quality decline in the short term (*Ontario Tree Marking Guide*, 2004). The ABCD system recognizes that log quality and vigor are separate considerations, and the classifications account for that. Trees are graded both on their ability to produce logs and their relative vigor or likelihood of declining in value. For example, a "Class A1" tree is defined as potentially capable of producing a high-quality log and maintaining that present quality for a 20-year

period. A “Class A” tree is identified as capable of producing a high-quality log, but the tree is identified as high-risk and likely to decline in quality within a 20-year period. The marking guide does not provide a system for identifying the volume of sawlog material proportional to merchantable volume in the stem.

Attempts have been made to create statistical models for predicting the likelihood of a tree producing sawlogs and the sawlogs’ grade (Bédard et al., 2018; Fortin et al., 2009). Two proposed classification systems that attempt to predict sawlog potential are “TG” and “hybrid RP.” TG involves classifying a tree into one of four grades based on external defects of the best 3.7 m portion of the bottom 5 m of the tree’s stem. The grade is assigned based on the length of the defect-free portion of the third-best face of the stem. Hybrid RP makes a binary classification of the sawlog potential in the standing tree and a binary assessment of a tree’s mortality risk (Bédard et al., 2018). Additional methods have been proposed for classifying stem quality in single species (Cecil-Cockwell & Caspersen, 2015). While these methods are of interest, to our knowledge none have been widely adopted in the field.

Assessing Tree Level Merchantability

The variability in quality found among hardwood stems makes it very difficult to predict sawlog volume (Cockwell & Caspersen, 2014). Castle et al. used an analysis of stem form and risk designed by the Northern Hardwood Research Institute to predict product potential in four hardwood species in the northern hardwood forest. Castle used a two-step approach, first modeling the likelihood of presence and then modeling the percentage of sawlog product present in trees that did contain a sawlog. The system developed by the Northern Hardwood Research Institute is also the system used for analysis in this study (Pelletier et al., 2016b). Castle condensed the original eight form classes into four groups and reduced the four risk classes to two, i.e., low risk and high risk. Castle found the four stem form classes, two risk classes, the logarithm of diameter at breast height (DBH), and species-specific DBH coefficients to be significant predictors of sawlog presence and volume in red oak, sugar maple, red maple, and yellow birch (Castle et al., 2017). Castle also found that simulations of stand value that included stem

form and risk as predictors in hardwood-dominant stands significantly differed from simulations that excluded form and risk as predictors. These differences ranged from 15.8% in stands composed of 50-75% hardwoods to 14.2% in stands composed of 75-100% hardwoods (Frank et al., 2018).

This analysis will assess the diversity of stem forms and risk found in northern hardwood forests representing a wide range of stand- and site-level conditions. The primary research objective is to evaluate the relationship between tree-, stand-, and site-level factors on stem form, risk, and merchantability across the common species in the region. In order to achieve this objective, this analysis will: (1) compare the occurrence of stem form and risk classifications for a given tree size across species; (2) relate stem form and risk to merchantability after accounting for additional factors; and (3) evaluate the robustness of the developed statistical equations for predicting tree merchantability.

Methods

Study Area

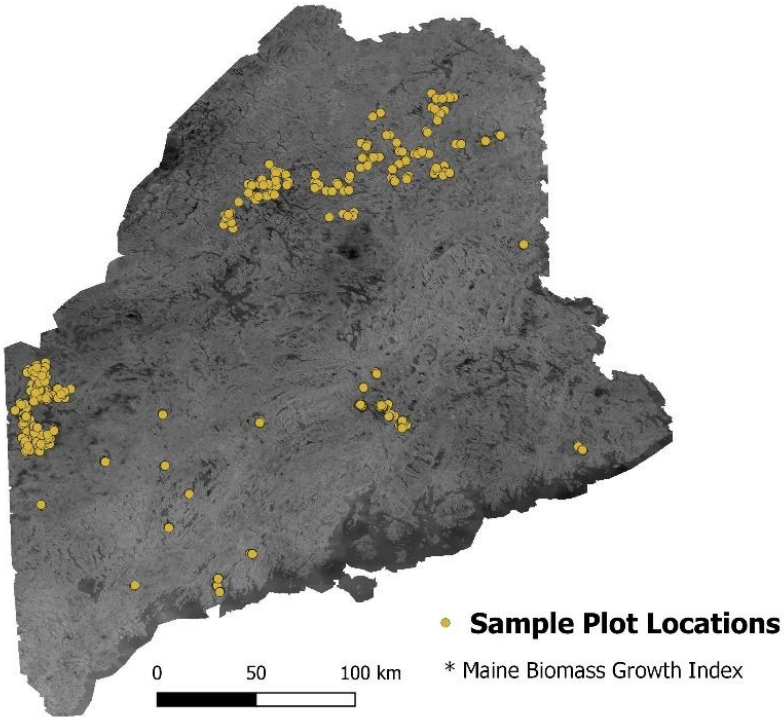


Figure 1.1 Sample plot locations used in this study set against the Maine biomass growth index.

The study was conducted in the Acadian Forest Region and consisted of 1,794 plots across the state of Maine (Figure 1.1). The species composition of the Acadian Region includes an abundance of softwood species such as red spruce and balsam fir. It also includes a large number of shade-tolerant hardwood species including red maple, sugar maple, American beech, and yellow birch, as well as intolerant hardwood

species such as paper birch, grey birch, and quaking aspen (Seymour, 1995). Eastern white pine, eastern hemlock, and northern red oak are also found in the southwestern portion of the region (Eyre, 1980) . Soils are classified as Spodosols, and while a number of factors influence species distribution, soil drainage is a key component (Fisher & Binkley, 2000). Soils primarily consist of podzols as a result of heavy rainfall and cool temperatures and can vary from acidic sandy loam to shallow and stony soils (Loo & Ives, 2003). Annual precipitation can vary widely across time and regions within the state, with annual precipitation ranging from 60 to 160 cm per year over the past two decades (Birkel, 2021). Regional topography is varied, with elevations ranging from sea level to 1,600 m.

Data Collection

Data for this study was obtained from four sources, all of which used the same methodology to identify form and risk as defined by the Northeastern Hardwood Research Institute (Castle et al., 2017; Pelletier et al., 2016b; *The Leaflet*, 2020). A total of five cruising teams collected data on tree diameter, species, form, and risk, measuring trees that had a diameter at breast height (DBH) greater than 10 cm. Each team conducted height (HT) measurements and field estimates of merchantable sawlog volume on a subsample of trees.

Each cruising team used the same definition of “sawlog material.” This definition required that trees meet species-specific minimum sawlog diameter requirements at breast height and that they contain at least two clear faces with no disqualifying defect such as excessive sweep, fruiting bodies, visible rot, large open cavities, or large limbs on at least one 2.43m (eight-foot) section of the stem. Minimum merchantable diameters for sawlogs were 25.4 cm for hardwoods, 12.7 cm for spruce and fir, and 20.3 cm for other softwood species. Four of the cruising teams measured the height of the sawlog portion of the stem with a hypsometer (Haglof Vertex 3) with the fifth making visual estimations of the length of the sawlog portion of the stem.

A total of 1,794 plots were measured, yielding a sample size of 18,003 trees. The sample was designed to capture the variable site conditions present in Maine by covering a large geography and a wide range of contrasting conditions across the state. All plots were conducted using variable radius sampling with the exception of one team that used .1 ha fixed-area plots. This sample was used both to examine the distribution of form and risk among species and to develop predictive models for form and risk. Of the total 18,003 trees in the sample, 3,908 trees (21.7%) were measured for sawlog volume. This sub-sample was used to develop predictive models for identifying the presence or absence of sawlog material within a tree. Of the trees where sawlog volume was estimated, 1,690 (43.2%) were determined to contain sawlogs. This sub-sample was used to develop equations for the estimated amount of merchantable volume composed of sawlog material in trees where a sawlog is present.

Variable Selection:

A large number of tree-level, one-sided competition, and two-sided competition metrics were assessed for significance. Additional site-specific conditions were assessed including climate, soil, slope and elevation. A complete list of variables tested for significance can be found in Table 1.1.

Table 1.1 Variables tested for significance in construction of predictive models of stem form, stem risk, sawlog presence, and percentage of merchantable volume avaliable as sawlog material.

Variable Type	**Variable**
Tree Level Measurements	Diameter at Breast Height (cm), Height (m), Height to Diameter Ratio (HT/DBH), Species, Stem Form, Stem Risk, Shade Tolerance, Standard Gravity, Drought Tolerance, Flood Tolerance
Climate Variables	26 Climate Variables Available Through ClimateNA (Appendix A), Climate Site Index
Soil Variables	Rooting Depth, Soil Drainage, Depth to Water Table, Soil Organic Carbon
Additional Site Characteristics	Percentage of Stand as Hardwood by Volume, QMD (cm), Canopy Height (m), Slope, Elevation (m), Biomass Growth Index, Maximum Stand Density Index, Mean Plot Shade Tolerance, Mean Plot Standard Gravity, Mean Plot Drought Tolerance, Mean Plot Flood Tolerance
Two-Sided Competition Metrics	Stand Density Index, Crown Competition Factor, Relative Density, Basal Area Per Hectare (m^2), Volume Per Hectare (m^3), Trees Per Hectare
One-Sided Competition Metrics	Crown Competition in Larger Trees, Basal Area in Larger Trees (m^2), Height to Canopy Ratio (m), Basal Area in Larger Hardwood Trees

Defining Stem Form and Risk

The NHRI system is composed of two components - form and risk (Table 1.2, Figure 1.2). Form refers to the shape of a tree's main stem based on its straightness, the presence and location of forks, inclination of the stem from the main axis, and the presence of multiple stems. These characteristics are consolidated into eight form classes (F1-F8). Risk is a measure of health and vigor, but it also considers damage to the stem that affects merchantability. Risk is defined by the tree's crown ratio, relative competition within the site, the presence of forks or multiple stems, defects on the main stem (such as burls), animal or mechanical damage, signs of decay, splits, holes, or fruiting bodies. Risk is consolidated into four categories that correlate to the types and severity of defect (R1-R4) (Pelletier et al., 2016b; *The Leaflet*, 2020). Two decision trees were developed by the NHRI to classify trees into form and risk classes based on the listed criteria, and these assisted in the classification of trees (Pelletier et al., 2016b).

Table 1.2. Description of NHRI form and risk classifications (Castle et al., 2018).

Classification	Description	Classification	Description
Stem form classification			
F1 	Ideal form; single straight stem; no significant fork on the first 5 m of the stem	F5 	Poor form; multiple stems are present between 0.3 and 1.3 m of the stem
F2 	Acceptable form; sweep on two axes or one significant curve on the stem	F6 	Poor form; single stem on the first 5 m; significant lean >15° on the vertical axis
F3 	Poor form; large branches on the first 5 m; large branches may have roundwood products	F7 	Acceptable form; significant fork between 2.5 and 5 m of the stem; large branches may have roundwood products
F4 	Unacceptable form; multiple stems or branches on the first 5 m; large branches do not have roundwood products	F8 	Poor form. Multiple stems present on the first 0.3 m of the stem
Stem risk classification			
R1 — low risk	Potential defects: crook, sweep, white face scar, burl	R3 — high risk	Potential defects: same as R1 and R2, cracks, open wounds, fruiting bodies, reduced crown ratio, dead branches
R2 — moderate risk	Potential defects: same as R1, small dark face scar, fire scar, animal damage, suppression	R4 — very high risk	Potential defects: same as R1, R2, and R3, multiple severe wounds (those included in R3)

Figure 3. Determination key for determining risk of losing vigor ("R" rating)

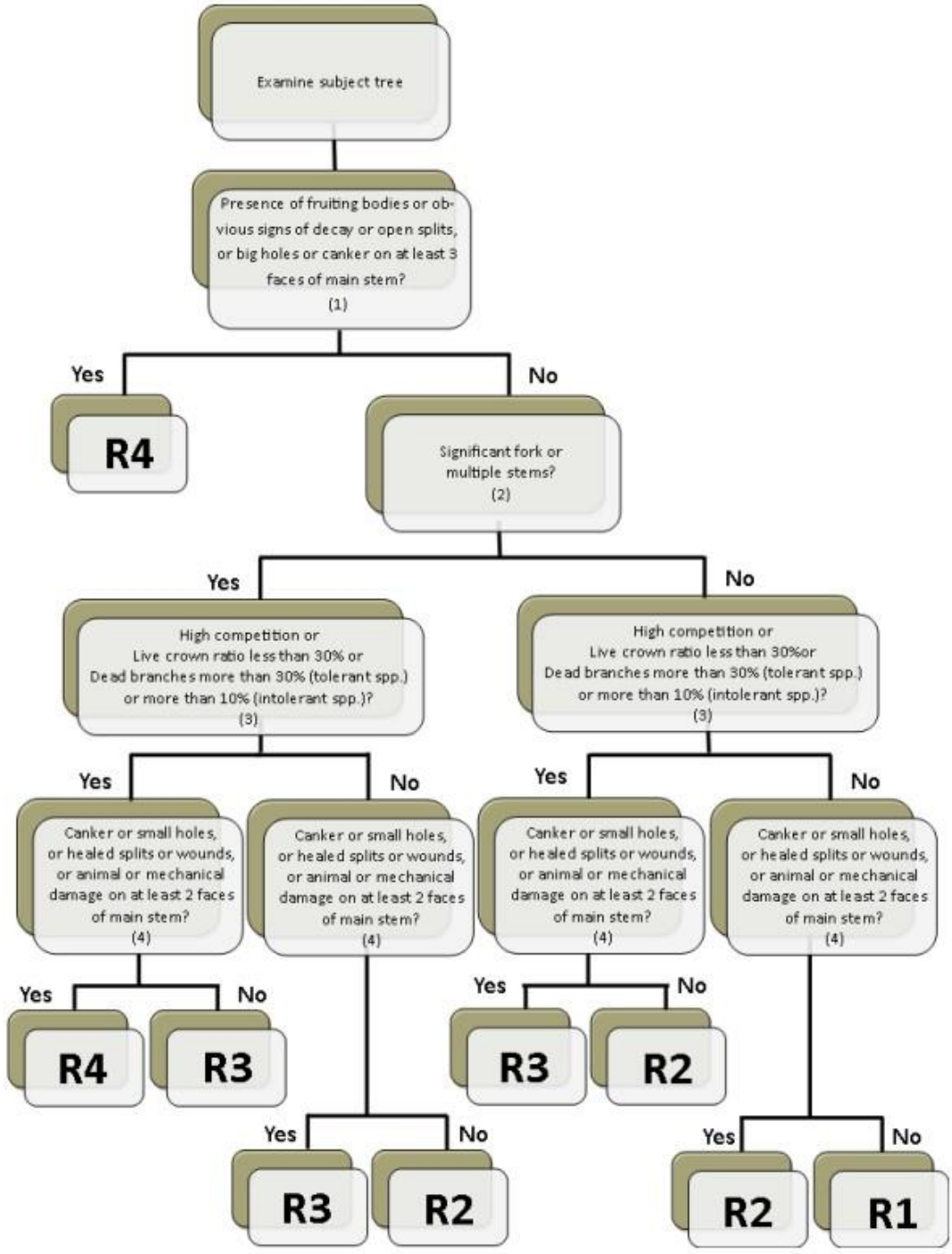


Figure 1.2 NHRI Risk Class Decision Tree (Pelletier et al., 2016b).

Statistical Methods

Consolidation of Form and Risk Classes

Redundancies existed between characteristics captured by different form and risk classes, so a least square means analysis was conducted to verify statistically significant differences in form and risk classes in predicting both sawlog likelihood and the ratio of sawlog volume to merchantable volume (Lenth, 2022). A mixed effect Analysis of Covariance (ANCOVA) test was conducted to assess differences in group means between form and risk classes. A mixed model was used to account for both the fixed effects of form and risk and random effects associated with variability among stands and plots within stands (Pinheiro, 2022).

Least square means were calculated with DBH, log(HT/DBH), and the basal area of larger trees included as covariates, as all were demonstrated to have a strong influence on sawlog presence and the ratio of sawlog volume to merchantable volume in trees that possessed sawlogs. The results of this analysis led to two form groups being created: ideal form (F1) and acceptable form (F2-F8). The analysis also led to the consolidation of risk into three groups: R1, R2, and R3 (which was composed of R3 and R4 observations).

Preliminary analysis also demonstrated significant differences in the distribution of form, risk, sawlog presence, and percentage of merchantable volume among hardwood species, spruce-fir species, and other sampled softwoods. For the remainder of the analysis, the sample was divided into these three groups.

Assessment of Form and Risk

Tree-specific, one-sided and two-sided competition, climate, site, and soil conditions were tested to assess the probability of ideal stem form and all three risk classes for hardwood, spruce-fir, and other softwood samples (Table 1.1).

Both the presence of ideal stem form and each risk class are binary, so a generalized linear model with a logit link was used to analyze the response. A mixed-effect framework was used to account for variability among stands and plots. The VSURF package in R (Genuer et al., 2019; R Core Team, 2021) was used to assist with variable selection.

Once potentially significant variables were identified for each response for the three species groups, the Akaike information criterion (AIC) was used to evaluate if a variable should be included in each model. Goodness of fit was determined by calculating the area under a receiver operator curve, which is a plot of sensitivity versus 1 minus specificity using the OptimalCutpoints and AUC packages in R both with and without random effects (Ballings & Van den Poel, 2013; Lopez-Raton & Rodriguez-Alvarez, 2021). AUC scores range from 0.5 to 1.0, where higher scores indicate a model with a greater capacity for discriminating true positive test values. Models with AUC scores between 0.5 and 0.7 have poor discrimination; those with scores between 0.7 and 0.8 have acceptable discrimination, and those with scores greater than 0.8 are considered to have excellent discrimination (Hosmer et al., 2013).

Table 1.3 Description of significant explanatory variables for predicting stem form, stem risk, the probability of sawlog presence, and percentage of merchantable volume composed of sawlog quality material.

Abbreviation	Definition	Source
DBH	Diameter at Breast Height (cm)	Field Measurement
HT	Height (m)	Field Measurement
I(HT/DBH)	Height (m) divided by Diameter at Breast Height (cm)	Field Measurement
Tallest	Plot level canopy height defined as the 100 tallest trees (m)	Field Measurement
BAL	Basal Area of Larger Trees (m^2)	Field Measurement
DD5	Degree-days above 5°C (growing degree days)	ClimateNA (Wang, 2015)
FFP	Frost-free period	(Wang, 2015)
MWMT	Mean temperature of the warmest month (°C)	ClimateNA (Wang, 2015)
MAP	Mean annual precipitation (mm)	ClimateNA (Wang, 2015)
BMGI	Biomass Growth Index	(Hennigar et al., 2017)
CMD	Hargreave's climatic moisture index	ClimateNA (Wang, 2015)
Eref	Hargreave's reference evaporation	ClimateNA (Wang, 2015)
Form	Form Class	NHRI Form Class (Castle et al., 2018)
Risk	Risk Class	NHRI Form Class (Castle et al., 2018)
CSI	Climate Site Index	(Jiang et al., 2015)
SPP	Species	Field Measurement

Form and Risk Prediction Models

The following models were developed for predicting ideal stem form, R1, R2, and R3 in hardwood species, spruce-fir, and other softwoods:

Hardwoods Form and Risk Models

[1]

$$pr(Form_{Ideal}) = \frac{\exp(\beta_0 + \beta_1 HT + \beta_2 DD5 + \beta_3 SPP)}{1 + \exp(\beta_0 + \beta_1 HT + \beta_2 DD5 + \beta_3 SPP)}$$

[2]

(2)

$$pr(Risk_1) = \frac{\exp(\beta_{20} + \beta_{21}(HT/DBH) + \beta_{22} FFP + \beta_{23} SPP)}{1 + \exp(\beta_{20} + \beta_{21}(HT/DBH) + \beta_{22} FFP + \beta_{23} SPP)}$$

[3]

$$pr(Risk_2) = \frac{\exp(\beta_{30} + \beta_{31} BAL + \beta_{32} FFP + \beta_{33} SPP)}{1 + \exp(\beta_{30} + \beta_{31} BAL + \beta_{32} FFP + \beta_{33} SPP)}$$

[4]

$$pr(Risk_3) = \frac{\exp(\beta_{40} + \beta_{41}log(HT/DBH) + \beta_{42}(Tallest/HT) + \beta_{43}BAL + \beta_{44}SPP)}{1 + \exp(\beta_{40} + \beta_{41}log(HT/DBH) + \beta_{42}(Tallest/HT) + \beta_{43}BAL + \beta_{44}SPP)}$$

See Table 1.3 for definitions of explanatory variables.

Spruce-Fir Form and Risk Models

[5]

$$pr(Form_{Ideal}) = \frac{\exp(\beta_{50} + \beta_{51}DBH + \beta_{52}MWMT + \beta_{53}MAP + \beta_{54}SPP + \beta_{55}(SPPxDBH))}{1 + \exp(\beta_{50} + \beta_{51}DBH + \beta_{52}MWMT + \beta_{53}MAP + \beta_{54}SPP + \beta_{55}(SPPxDBH))}$$

[6]

$$pr(Risk_1) = \frac{\exp(\beta_{60} + \beta_{61}log(DBH) + \beta_{62}MWMT + \beta_{63}MAP + \beta_{64}SPP)}{1 + \exp(\beta_{60} + \beta_{61}log(DBH) + \beta_{62}MWMT + \beta_{63}MAP + \beta_{64}SPP)}$$

[7]

$$pr(Risk_2) = \frac{\exp(\beta_{70} + \beta_{71}log(DBH) + \beta_{72}MAP + \beta_{73}DD5)}{1 + \exp(\beta_{70} + \beta_{71}log(DBH) + \beta_{72}MAP + \beta_{73}DD5)}$$

[8]

$$pr(Risk_3) = \frac{\exp(\beta_{80} + \beta_{81}DD5 + \beta_{82}CMD + \beta_{83}SPP)}{1 + \exp(\beta_{80} + \beta_{81}DD5 + \beta_{82}CMD + \beta_{83}SPP)}$$

See Table 1.3 for definitions of explanatory variables.

Other Softwoods Form and Risk Models

[9]

$$pr(Form_{Ideal}) = \frac{\exp(\beta_{90} + \beta_{91}log(HT/DBH) + \beta_{92}Eref + \beta_{93}SPP)}{1 + \exp(\beta_{90} + \beta_{91}log(HT/DBH) + \beta_{92}Eref + \beta_{93}SPP)}$$

[10]

$$pr(Risk_1) = \frac{\exp(\beta_{100} + \beta_{101}log(DBH) + \beta_{102}log(Tallest) + \beta_{103}MWMT + \beta_{104}SPP)}{1 + \exp(\beta_{100} + \beta_{101}log(DBH) + \beta_{102}log(Tallest) + +\beta_{103}MWMT + \beta_{104}SPP)}$$

[11]

$$pr(Risk_2) = \frac{\exp(\beta_{110} + \beta_{111}log(DBH) + \beta_{112}BMGI + \beta_{113}SPP + \beta_{114}SPPxlog(DBH))}{1 + \exp(\beta_{110} + \beta_{111}log(DBH) + \beta_{112}BMGI + \beta_{113}SPP + \beta_{114}SPPxlog(DBH))}$$

[12]

$$pr(Risk_3) = \frac{\exp(\beta_{120} + \beta_{121} log(DBH) + \beta_{122} BAL + \beta_{123} BMGI + \beta_{124} SPP)}{1 + \exp(\beta_{120} + \beta_{121} log(DBH) + \beta_{122} BAL + \beta_{123} BMGI + \beta_{124} SPP)}$$

See Table 1.3 for definitions of explanatory variables.

Sawlog Recovery

The assessment of sawlog recovery was analyzed in two stages with the methodology used by Castle et al. (2017) in their previous assessment of sawlog recovery using form and risk class. First, the presence of at least one sawlog in a tree was treated as a binary event, so a generalized linear model with a logit link was used to predict the presence of a sawlog. The same variables and statistical methods utilized for the analysis above were used for the development of sawlog likelihood models. The following three models were developed for each specific species group:

Sawlog Presence Models

Hardwoods

[13]

$$pr(Sawlog) = \frac{\exp(\beta_{131} + \beta_{132} log(DBH) + \beta_{133} ln(HT/DBH) + \beta_{134} BAL + \beta_{135} Form + \beta_{136} Risk + \beta_{137} (SPPxDBH))}{1 + \exp(\beta_{131} + \beta_{132} log(DBH) + \beta_{133} ln(HT/DBH) + \beta_{134} BAL + \beta_{135} Form + \beta_{136} Risk + \beta_{137} (SPPxDBH))}$$

Spruce-fir

[14]

$$pr(Sawlog) = \frac{\exp(\beta_{140} + \beta_{141} log(DBH) + \beta_{142} Risk + \beta_{143} log(HT) + \beta_{144} (SPPxDBH))}{1 + \exp(\beta_{140} + \beta_{141} log(DBH) + \beta_{142} Risk + \beta_{143} log(HT) + \beta_{144} (SPPxDBH))}$$

Other Softwood

[15]

$$pr(Sawlog) = \frac{\exp(\beta_{150} + \beta_{151} log(DBH) + \beta_{152} Risk + \beta_{153} CSI + \beta_{154} (SPPxDBH) + \beta_{155} (SPPxHT))}{1 + \exp(\beta_{150} + \beta_{151} log(DBH) + \beta_{152} Risk + \beta_{153} CSI + \beta_{154} (SPPxDBH) + \beta_{155} (SPPxHT))}$$

See Table 1.3 for definitions of explanatory variables.

Percentage of Merchantable Volume as Sawlog Quality

The second step was to develop models predicting the proportion of merchantable volume in each tree containing a sawlog that was composed of sawlog material. For trees with no measured heights, a species- and plot-specific modified mixed-effects model was utilized, similar in nature to a prior study (A. P. Robinson & Wykoff, 2004). Regional Kozak's taper equations were used for all trees in order to estimate both total merchantable volume and the field measured sawlog volume for each tree determined to be of sawlog quality by the field crews (Kozak, 2004; Li et al., 2012; Li & Weiskittel, 2010). Sawlog volume was calculated using the heights of the sawlog portion of each stem measured in the field.

Once these volume calculations were made, the samples were divided again into hardwoods, spruce-fir, and other softwoods. The proportion of sawlog to merchantable volume was constrained between zero and one, so beta regression was chosen to model the response and a logit link was selected to establish a linear relationship with the predictor variables (Castle et al., 2017). Along with the consolidated form and risk classes, all of the previously mentioned tree-level, competition, and site-level variables were tested for significance. The following models were developed to predict the proportion of sawlog to merchantable volume:

Hardwoods

[16]

$$S_{vol}/M_{vol} = \frac{\exp(\beta_{161} + \beta_{162}log(DBH) + \beta_{163}ln(HT/DBH) + \beta_{164}Form + \beta_{165}Risk + \beta_{166}(SPPxDBH))}{1 + \exp(\beta_{161} + \beta_{162}log(DBH) + \beta_{163}ln(HT/DBH) + \beta_{164}Form + \beta_{165}Risk + \beta_{166}(SPPxDBH))}$$

Spruce-fir

[17]

$$S_{vol}/M_{vol} = \frac{\exp(\beta_{170} + \beta_{171}log(DBH) + \beta_{172}Form + \beta_{173}(SPPxDBH))}{1 + \exp(\beta_{170} + \beta_{171}log(DBH) + \beta_{172}Form + \beta_{173}(SPPxDBH))}$$

Other Softwoods

[18]

$$S_{vol}/M_{vol} = \frac{\exp(\beta_{180} + \beta_{181} log(DBH) + \beta_{182} Risk + \beta_{183} CSI + \beta_{184}(SPPxDBH))}{1 + \exp(\beta_{180} + \beta_{181} log(DBH) + \beta_{182} Risk + \beta_{183} CSI + \beta_{184}(SPPxDBH))}$$

See Table 1.3 for definitions of explanatory variables.

Results

Stem Form and Risk

Hardwood Species

Red maple and sugar maple had the largest sample sizes among hardwood tree species sampled, followed by yellow birch, red oak, paper birch, quaking aspen, American beech, and white ash (Table A.9). All other hardwood species had fewer than twenty samples. DBH measurements across species ranged from 10.2 to 94.7 cm with a mean diameter of 25.3cm ± 12.3cm.

A total of 8,674 hardwood stems were utilized to predict stem form and risk, and 2,798 of those trees were merchandized (Table 1.4). In both samples, form and risk class were highly variable. Straight stems (F1) were the most observed form class comprising 45.3% of the larger sample group and 47.6% of the merchandized trees. Sweep on at least two axes (F2) and trees with multiple stems (F8) were the second and third most observed form classes. Risk class was also highly variable among hardwoods, and a similar distribution was found between the total sample of trees and those trees that were merchandized, with 46.2%, 34.7%, and 16.9% of the merchandized trees being observed as risk classes one, two, and three, respectively (Table 1.4). Additional summary tables can be found in Appendix A.

Table 1.4 Summary of tree-level attributes of species sampled for form and risk, merchandized, and sawlog samples. Definitions: DBH, diameter at breast height cm; HT, height m; BAL, basal area of larger trees m^2; SD, standard deviation; min, minimum measured value; max, maximum measured value

Attributes	Hardwoods	Spruce-Fir	Other Softwoods	Total
Total Trees (Total Sample)	**8674**	**5019**	**4310**	**18003**
DBH (Mean; SD)[min, max]	(25.3, 12.3)[10.2, 94.7]	(19.9, 8.6)[10.2, 74.7]	(29.0, 13.9)[10.2, 110.0]	(24.7, 12.3)[10.2, 110.0]
HT (Mean; SD)[min, max]	(17.9,3.4) [4.9, 37.8]	(15.0, 2.9) [5.5, 31.4]	(18.7, 4.3) [5.8, 34.7]	(17.3, 3.8) [4.9, 37.8]
BAL(Mean;SD)[min, max]	(15.8, 12.2) [0, 68.9]	(15.0, 12.1) [0, 73.5]	(17.1, 14.1) [0, 73.5]	(15.9, 12.7) [0, 73.5]
Risk Class				
1 (%)	4106 (47.3)	3446 (68.7)	2183 (50.6)	9735 (54.1)
2 (%)	3107 (35.8)	1208 (24.1)	1457 (33.8)	5772 (32.1)
3 (%)	1461 (16.9)	365 (7.2)	670 (15.6)	2496 (13.8)
Form Class				
Ideal (%)	3935 (45.3)	4858 (96.8)	3438 (79.8)	12231 (67.9)
Acceptable (%)	4739 (54.7)	161 (3.2)	872 (20.2)	5772 (32.1)
Total Trees (Merchandized)	**2798**	**818**	**254**	**3870**
Has Sawlog (%)	828 (29.6)	690 (84.4)	164 (64.8)	1682 (43.5)
DBH (Mean; SD)[min, max]	(27.8, 13.3)[10.2, 92.7]	(22.2, 9.0)[11.7, 56.9]	(38.6, 15.6)[11.9, 110.0]	(27.3, 13.3)[10.2, 110.0]
HT (Mean; SD)[min, max]	(18.5, 3.8) [4.9, 37.8]	(15.0, 3.6) [5.5, 31.4]	(18.6, 5.4) [9.4, 34.7]	(17.7, 4.1) [4.9, 37.8]
BAL(Mean; SD)[min, max]	(12.7, 10.2) [0, 59.7]	(15.2, 12.0) [0, 64.3]	(11.0, 13.9) [0, 73.5]	(13.1, 10.9) [0, 73.5]
Risk Class				
1 (%)	1292 (46.2)	726 (88.8)	186 (73.2)	2204 (57.0)
2 (%)	971 (34.7)	42 (5.1)	34 (13.4)	1047 (27.0)
3 (%)	535 (19.1)	50 (6.1)	34 (13.4)	619 (16.0)
Form Class				
Ideal (%)	1332 (47.6)	803 (98.1)	208 (81.9)	2343 (60.5)
Acceptable (%)	1466 (52.4)	15 (1.9)	46 (18.1)	1527 (39.5)
Total Trees (Has Sawlog)	**828**	**690**	**164**	**1682**
Svol/Mvol (Mean, SD)	(53.3, 18.5)	(84.4, 16.6)	(77.5, 21.1)	(68.4, 23.5)
DBH (Mean, SD)[min, max]	(38.6, 9.8)[25.7, 92.7]	(23.2, 8.8)[13.0, 56.9]	(41.4, 13.3)[21.8, 82.3]	(32.5, 12.6)[13.0, 92.7]
HT (Mean, SD)[min, max]	(20.4, 2.9) [9.8, 37.8]	(15.5, 3.5)[8.5, 31.4]	(20.4, 4.8) [9.4, 34.7]	(18.8, 4.4) [8.5, 37.8]
BAL(Mean, SD)[min, max]	(9.1, 7.6) [0, 36.7]	(14.1, 11.3) [0, 59.7]	(11.6, 14.8) [0, 68.9]	(11.4, 10.4) [0, 68.9]
Risk Class				
1 (%)	443 (53.5)	651 (94.3)	122 (74.4)	1216 (72.3)
2 (%)	329 (39.7)	28 (4.1)	26 (15.9)	383 (22.8)
3 (%)	56 (6.8)	11 (1.6)	16 (9.7)	83 (4.9)
Form Class				
Ideal (%)	544 (65.7)	684 (99.1)	134 (81.7)	1362 (81.0)
Acceptable (%)	284 (34.3)	6 (0.9)	30 (18.3)	320 (19.0)

Of the 2,798 hardwood trees that were merchandized, almost 30% had at least one sawlog present. The most common tree species in this sample were sugar maple followed by yellow birch and red oak. Hardwood trees with sawlogs were more likely to be of ideal form (65.7%) and low risk (53.5%) than the trees in the entire merchandized sample. Only 6.8% of the trees that contained sawlogs were considered high risk, compared with 19.1% in the merchandized sample overall. The mean DBH increased to 38.6 ± 9.8 cm for trees with sawlogs from 27.8 ± 13.3 cm in the total sample of trees that were merchandized. The mean HT increased from 18.5 ± 3.8 m to 20.4 ± 2.9 m. The variance in DBH and HT in the sample that contained sawlogs decreased when compared to the total sample of merchandized trees (Table 1.4).

Using only fixed effects, models developed to predict ideal stem form and all three significant risk classes (R1, R2, and R3) in hardwoods each demonstrated relatively limited discrimination. The models had AUC values of 0.65, 0.66, 0.65, and 0.63, respectively, for the models without random effects. Tree-level variables were significant in predicting ideal form class ($p < .001$). Tree-level and one-sided competition variables demonstrated significance in the R2 and R3 models ($p < .001$). Site-level climate variables were significant in predicting ideal stem form and risk classes R1 and R2 ($p <.001$). The number of growing days (degree-days above 5°C) had a negative and significant relationship with a hardwood stem demonstrating ideal form. The number of frost-free days had a negative relationship with risk class R1 but a positive relationship with risk class R2. See Appendix A for model parameter estimates along with standard errors and goodness of fit statistics.

Spruce-fir

Preliminary analysis demonstrated no significant difference between form, risk, sawlog presence, or sawlog recovery between red and white spruce species, so these observations were grouped together for the rest of the analysis ($p <.05$). Black spruce was included with other softwoods due to the relatively small sample sizes and its differences in preferred site qualities from the other

spruce-fir species (Farrar, 1995). Of the sample of 5,019 total spruce-fir trees used for assessing stem form and risk, the minimum diameter was 11.7 cm, the maximum diameter was 74.7 cm, and the mean diameter was 19.9 ± 8.6 cm. There was little variability in form class distribution, with 96.8% of samples demonstrating a straight stem with no forking below 5 m. Risk class was more variable with 68.7%, 24.1%, and 7.2% of samples identified as risk classes R1, R2, and R3, respectively.

A total of 818 spruce-fir trees were merchandized with 84.4% containing sawlogs. In the sample that was merchandized, ideal stem form represented 98.1% of the sample, and this increased to 99.1% in the trees with sawlogs. Risk class was less variable in the trees that were merchandized with 88.8% of trees in risk class R1. This increased to 94.3% in the trees with sawlogs. The mean diameter of spruce and fir trees that were merchandized was 22.2 ± 9.0 cm. The spruce-fir trees with sawlogs demonstrated a similar mean diameter and variance with a mean of 23.2 ± 8.8 cm. A similar trend was seen with tree height (Table 1.4).

Unlike the models developed for hardwoods, using only fixed effects, the model developed to predict ideal stem form and risk class 3 in spruce and fir trees demonstrated acceptable discrimination with AUC values of 0.72 and 0.73, respectively. The models to predict risk classes R1 and R2 demonstrated excellent discrimination with AUC values of 0.89 and 0.88, respectively. Site-level variables constituted a majority of the significant variables for all four models. These included mean temperature of the warmest month for stem form ($p < .001$) and multiple site-level climate variables for all three risk class models including mean annual precipitation, growing degree days and Hargreave's climatic moisture index ($p < .01$). See Appendix A for model parameter estimates along with standard errors and goodness of fit statistics.

Other Softwoods

The sample of other softwoods contained 1,827 Eastern white pine, 668 Northern white-cedar, and 1,827 Eastern hemlock. All other softwood species contained fewer than 80 observations and were grouped with these other prior species in preliminary analysis. Of the sample of 4,310 other softwood trees, the DBH measurements ranged from 10.2 to 110.0 cm with a mean diameter of 29.0 ± 13.9 cm. There was less variability in stem form within this species group when compared to hardwoods with 79.8% of the trees displaying a straight stem with no forking below 5 m (F1). There was high variability in the risk class observations with 50.6%, 33.8%, and 15.6% of the sample trees identified as R1, R2 and R3, respectively.

Of the larger sample of 4,310 trees, 268 trees were merchandized. The minimum DBH for merchandized trees was 11.9 cm and the maximum was 110.0 cm with a mean of 38.6 ± 15.6. A single straight stem was observed in 81.9% of the merchandized trees and 73.2% were risk class 1. Of the merchandized trees, 64.8% contained sawlogs. The mean diameter increased to 41.4 ± 13.3 cm in the sample containing sawlogs. The mean height in the trees with sawlogs increased to 20.4 ± 4.8 m in the merchandized sample, as compared with a mean of 18.6 ± 5.4m in the overall merchandized sample (Table 1.4).

Using only fixed effects, the models predicting ideal stem form and risk class 2 in other softwoods demonstrated poor discrimination with AUC values of 0.65 and 0.68. Acceptable discrimination was demonstrated in the models predicting risk class 1 and risk class 3 with AUC values of 0.77 and 0.73, respectively. Tree-level variables, canopy height and metrics of competition demonstrated the majority of significant variables in the models for other softwoods ($p < .001$). A specific site-level variable was significant in each model with estimated evaporation being significant in the ideal stem form model and the mean temperature of the warmest month being significant in the risk class R1 model with p-values <0.001. The biomass growth index was significant in risk class R2 and risk

class R3 models with p-values of 0.019 and 0.002, respectively (see Appendix for model summaries along with standard errors and goodness of fit statistics).

Sawlog Presence and Recovery

Hardwood Species

Only tree-level and one-sided competition variables were significant for predicting sawlog presence in hardwood species ($p < .05$). The binomial model demonstrated excellent discrimination with an AUC value of 0.93 using only fixed effects. DBH had a nonlinear yet positive effect across species (Figure 1.3, Table 1.5). Sawlog likelihood varied across species, with red oak, quaking aspen and white ash showing higher likelihood of sawlog presence across diameter classes than sugar maple, paper birch and red maple (Figure 1.3). The influence of basal area in larger species showed a similar trend for these species as well (Figure 1.4).

Table 1.5 Parameter estimates with associated standard errors (SE) and fit statistics for equations 13 and 16.

	Eq. 13 (Binomial) Hardwoods Has Sawlog				Eq. 16 (beta) Hardwoods Svol/Mvol			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 131$	-37.4916	2.1573	<.001	$\beta 161$	-35.1440	2.1969	<.001
log(DBH)	$\beta 132$	12.6034	0.8745	<.001	$\beta 162$	12.7029	0.8555	<.001
log(HT/DBH)	$\beta 133$	2.3705	0.6924	<.001	$\beta 163$	-0.7946	0.1448	<.001
BAL	$\beta 134$	-0.0319	0.0145	.029				
Form1	$\beta 135$	1.2739	0.1877	<.001	$\beta 164$	0.1393	0.0395	<.001
Risk2	$\beta 136$	-0.7455	0.2097	<.001	$\beta 165$	-0.1801	0.0436	<.001
Risk3	$\beta 136$	-4.0545	0.2592	<.001	$\beta 165$	-0.2903	0.0796	<.001
AB X DBH	$\beta 137$	-0.1166	0.0406	.004				
OH x DBH	$\beta 137$	-0.1436	0.0347	<.001	$\beta 166$	-0.2864	0.0243	<.001
PB x DBH	$\beta 137$	-0.1331	0.0345	<.001	$\beta 166$	-0.2928	0.0242	<.001
QA x DBH	$\beta 137$	-0.1451	0.0350	<.001	$\beta 166$	-0.2823	0.0241	<.001
RM x DBH	$\beta 137$	-0.1522	0.0302	<.001	$\beta 166$	-0.2916	0.0240	<.001
RO x DBH	$\beta 137$	-0.0400	0.0311	.199	$\beta 166$	-0.2911	0.0240	<.001
SM x DBH	$\beta 137$	-0.1405	0.0276	<.001	$\beta 166$	-0.2901	0.0238	<.001
WA x DBH	$\beta 137$	-0.0960	0.0403	.017	$\beta 166$	-0.2791	0.0243	<.001
YB x DBH	$\beta 137$	-0.1414	0.0281	<.001	$\beta 166$	-0.2886	0.0238	<.001
Fit Statistics								
AUC (FE, RE)		(0.93, .97)	(.01, .01)			-		
R2		-	-			.29		
RMSE		-	-			0.1534		
MAE		-	-			0.1202		
Bias		-	-			-0.0068		

***Note:** Svol/Mvol, proportion of merchantable volume that is sawlog quality; Ind. variable, independent variable; Par., parameter; DBH, diameter at breast height; HT, height; BAL, basal area in larger trees; Form1, NHRI form class 1; Risk2, NHRI risk class 2; Risk3, NHRI risk classes 3 and 4; AB, american beech; OH, other hardwoods; PB, paper birch; QA, quaking aspen; RM, red maple; RO, red oak; SM, sugar maple; WA, white ash; YB, yellow birch; AUC, area under the curve; FE pertains to models with AUC derived from fixed effects only; RE pertains to models with AUC derived from both fixed and random effects. RMSE, root mean square error; MAE, mean actual error; NHRI form classes 2-8 and NHRI risk class 1 were used as reference levels.

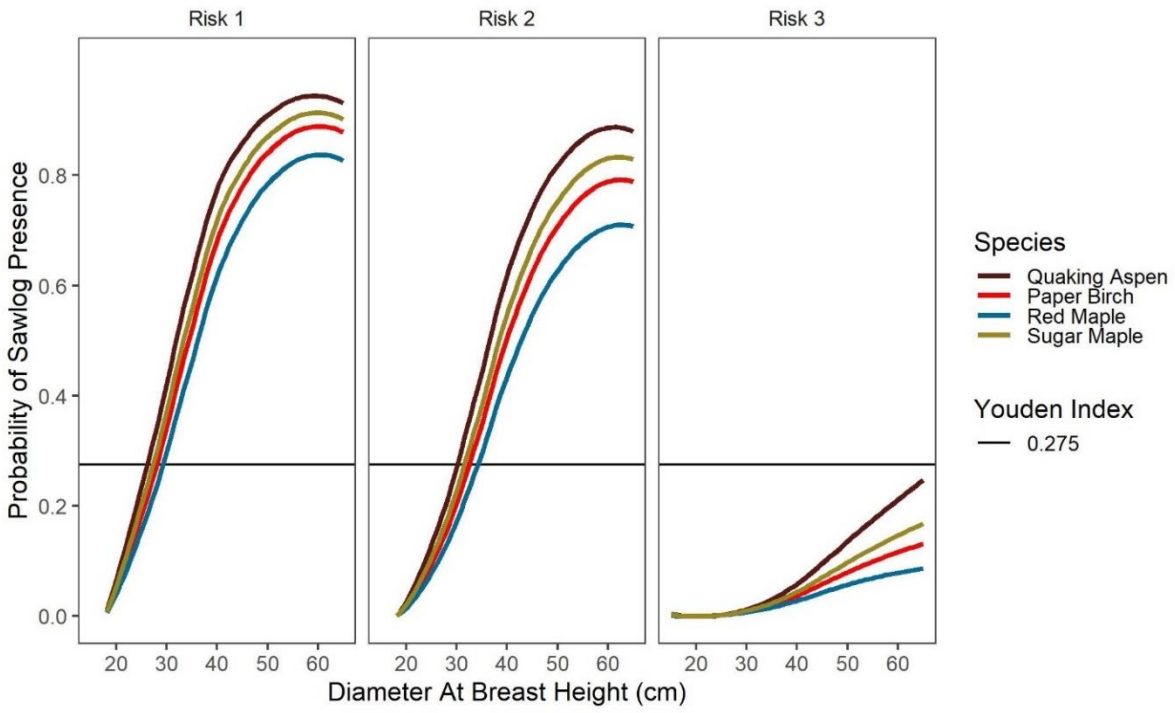


Figure 1.3 Effect of diameter on likelihood of sawlog presence across risk classes for four hardwood species. Figure represents model predictions for trees with a mean BAL of 15.8 m^2 and a height of 17.9 m. Youden index represents the calculated optimal threshold value for determination of sawlog presence (Fluss et al., 2005).

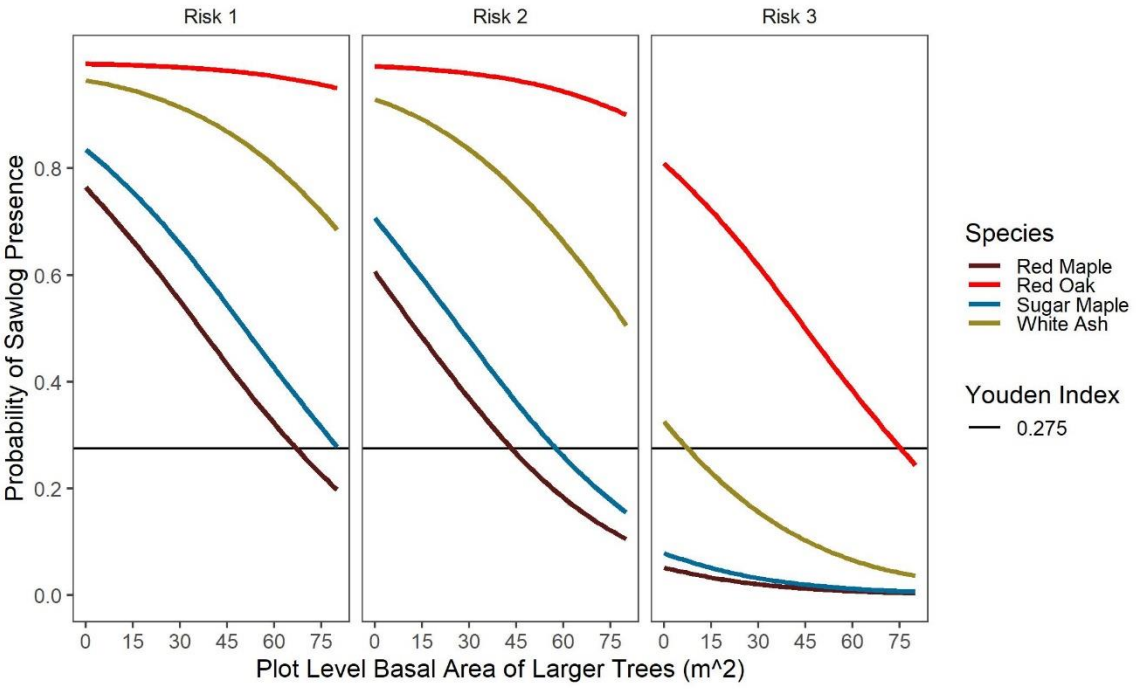


Figure 1.4 Effect of the basal area of larger trees on likelihood of sawlog presence across risk classes for four hardwood species. Figure represents model prediction for trees with a mean height of 20.4 m and a mean dbh of 38.25 cm. Youden index represents the calculated optimal threshold value for determination of sawlog presence (Fluss et al., 2005).

For predicting the percentage of merchantable volume available as sawlog material in the hardwood species tested, only tree-level variables were identified as significant ($p < .001$). Much like sawlog presence, DBH had a non-linear effect on sawlog recovery rates. However, unlike sawlog presence, there was a significant negative relationship between DBH and merchantable volume available as sawlog material for each species upon reaching a species-specific minimum DBH. The precision of the fixed-effect only predictions were generally low (RMSE = 0.15) with an R^2 of 29%, but it displayed little bias (-.0068) (Table 1.5).

When the binary and beta models were combined using a probabilistic approach, the predicted maximum percentage of merchantable volume available as sawlog material was seen to occur at a higher DBH as risk increased across species. In low-risk trees, sugar maple and yellow birch reached similar maximum values of sawlog material at approximately 66% of merchantable volume at a DBH of 51 cm, while white ash reached its maximum value of 81% at a DBH of 50 cm. Red maple had a lower maximum value of 54%, which was reached at a DBH of 51 cm. All species demonstrated decreasing ratios of sawlog material to merchantable volume after a DBH of 60 cm regardless of form or risk class (Figure 1.5).

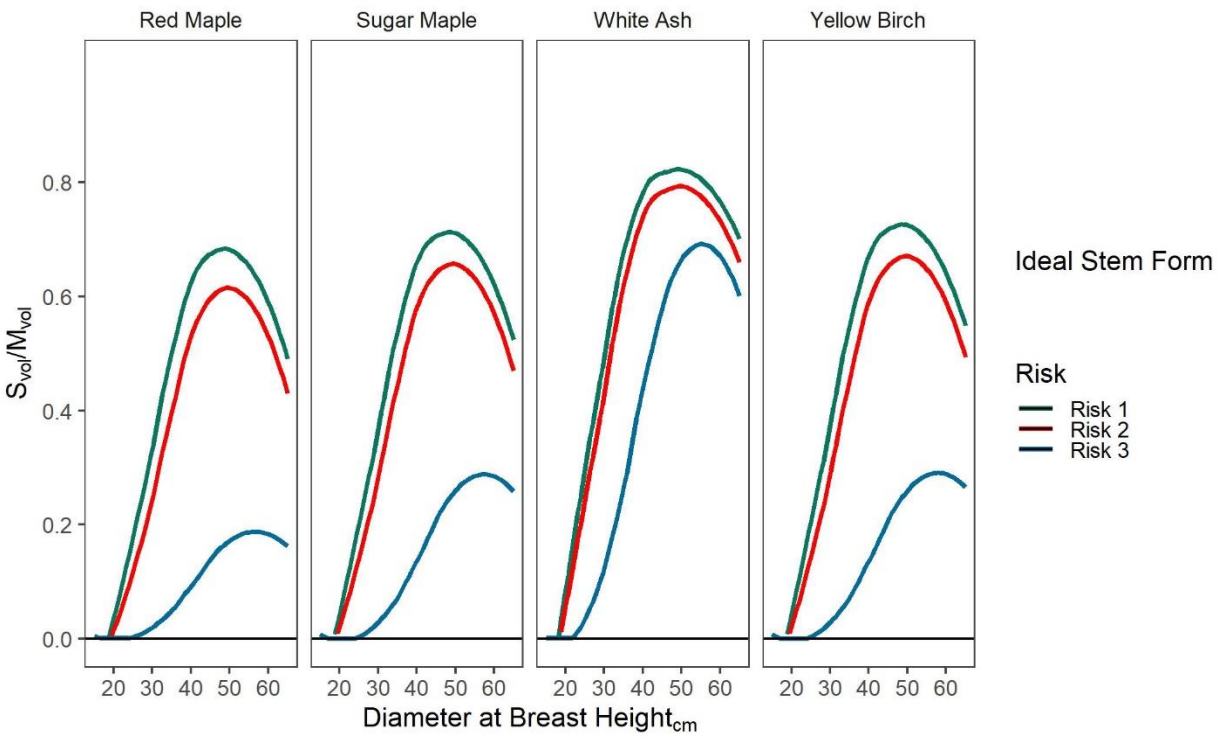


Figure 1.5 The percentage of merchantable volume composed of sawlog quality material for four hardwood species across diameter classes using a two-step probabilistic model that multiplies the predicted value of equations 13 and 16. Figure represent model predictions for a tree with a height of 18.5 m. S_{vol}/M_{vol} represents the proportion of merchantable volume that is predicted to be sawlog quality.

Spruce-Fir

When tested for significance, only tree-level variables were significant for predicting the presence of sawlogs ($p < .001$) (Table 1.6). The binomial model demonstrated excellent discrimination with an AUC value of 0.90 when considering only fixed effects. The relationship between sawlog presence and diameter was non-linear, with significantly decreased sawlog likelihood upon reaching a specific DBH that was dependent upon the tree risk class.

Only tree level variables were identified as significant for predicting the percentage of merchantable volume available as sawlog material for spruce-fir trees ($p < .003$). Much like sawlog presence, there was a significant negative relationship between DBH and merchantable volume available as sawlog material for each species upon reaching a species-specific DBH ($p < .001$). The precision of the model was high, and it displayed little bias (-.0034). The RMSE for the model was 0.061, and the R^2 was 0.78.

When the models were combined using a probabilistic approach, the predicted maximum percentage of merchantable volume available as sawlog material was above 90% in both risk classes R1 and R2 with maximum sawlog potential occurring at approximately 30 cm DBH. R1 trees maintained a ratio of sawlog material near 90% at 50 cm DBH, while R2 trees had a ratio of only 31% and 25% in spruce and fir trees, respectively, at 50 cm DBH. R3 trees had low ratios of sawlog to merchantable volume at all DBH values (Figure 1.6).

Table 1.6 Parameter estimates with associated standard errors (SE) and fit statistics for equations 14 and 15.

	Eq. 14 (Binomial) Spruce-Fir Has Sawlog				Eq. 15 (Binomial) Other Softwoods Has Sawlog			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 140$	-55.4585	4.3643	<.001	$\beta 150$	-79.3273	8.0457	<.001
log(DBH)	$\beta 141$	23.3889	2.1257	<.001	$\beta 151$	24.8516	3.6384	<.001
Risk2	$\beta 142$	-3.6702	0.6626	<.001	$\beta 152$	-1.4568	0.8860	.103
Risk3	$\beta 142$	-9.5663	0.6336	<.001	$\beta 152$	-4.6883	0.9626	<.001
log(HT)	$\beta 143$	3.2084	0.9879	.001				
CSI					$\beta 153$	0.3641	0.1442	.013
BF x DBH	$\beta 144$	-0.8407	0.0877	<.001				
RS x DBH	$\beta 144$	-0.8472	0.0835	<.001				
OS x DBH					$\beta 154$	-0.8246	0.1642	<.001
EH x DBH					$\beta 154$	-0.6503	0.1021	<.001
OC x DBH					$\beta 154$	-0.6121	0.1044	<.001
WP x DBH					$\beta 154$	-0.5534	0.0814	<.001
OS x log(HT)					$\beta 155$	6.2570	2.2138	.006
EH x log(HT)					$\beta 155$	4.1018	2.0928	.053
OC x log(HT)					$\beta 155$	2.9770	2.0957	.158
WP x log(HT)					$\beta 155$	3.4551	2.0391	.093
Fit Statistics								
AUC (FE, RE)		(.90, .94)	(.03, .02)			(.88, .97)	(.04, .02)	

*Note: Ind. variable, independent variable; Par., parameter; DBH, diameter at breast height; Risk2, NHRI risk class 2; Risk3, NHRI risk classes 3 and 4; HT, height; CSI, climate site index; BF, balsam fir; RS, red spruce; OS, other softwoods; EH, eastern hemlock; OC, northern white-cedar; WP, white pine; FE pertains to models with AUC derived from fixed effects only; RE pertains to models with AUC derived from both fixed and random effects. NHRI risk class 1 were used as reference levels.

Only tree level variables were identified as significant for predicting the percentage of merchantable volume available as sawlog material for spruce-fir trees ($p < .003$). As with sawlog presence, there was a significant negative relationship between DBH and merchantable volume available as sawlog material for each species upon reaching a species-specific DBH ($p < .001$). The precision of the model was high, and it displayed little bias (-.0034). The RMSE for the model was 0.061, and the R^2 was 0.78.

When the models were combined using a probabilistic approach, the predicted maximum percentage of merchantable volume available as sawlog material was above 90% in both risk classes R1 and R2 with maximum sawlog potential being reached at approximately 30 cm DBH. R1 trees maintained a ratio of sawlog material near 90% at 50 cm DBH, while R2 trees had a ratio of only 31% and 25% in spruce and fir trees, respectively, at 50 cm DBH. R3 trees had low ratios of sawlog to merchantable volume at all DBH values (Figure 1.6).

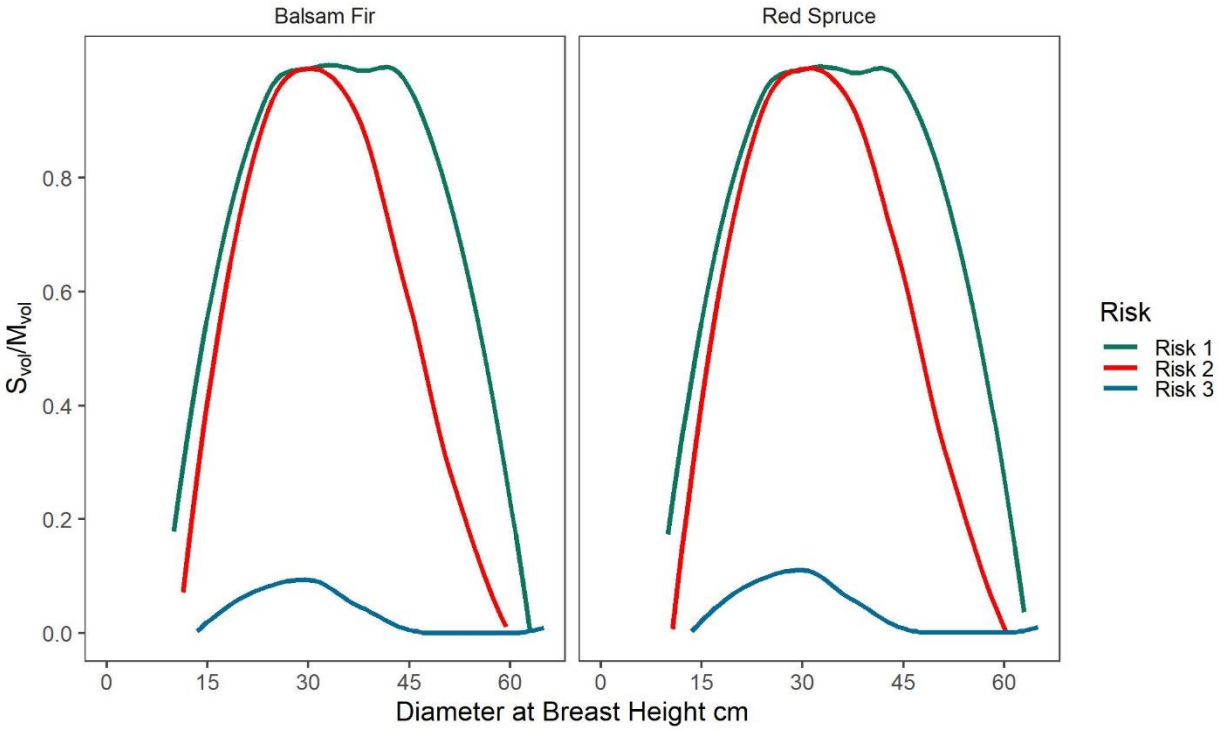


Figure 1.6 The percentage of merchantable volume composed of sawlog quality material for balsam fir and red spruce across diameter classes using a two-step probabilistic model that multiplies the predicted value of equations 14 and 17. Figure shows results for trees with a mean height of 15.5 m and an average of the values from stems with both ideal and acceptable stem forms. S_{vol}/M_{vol} represents the proportion of merchantable volume that is predicted to be sawlog quality.

Other Softwoods

When tested for significance, tree- and site-level variables were significant for predicting the presence of a sawlog in other softwood species tested ($p < .05$) (Table 1.6). The binomial model demonstrated excellent discrimination with an AUC value of 0.89 when considering only fixed effects. The relationship between sawlog presence and DBH was non-linear with a significant decrease in sawlog likelihood upon reaching a specific DBH that was dependent upon the tree species and risk class (Figure 1.7).

In addition, tree- and site-level variables were identified as significant for predicting the percentage of merchantable volume available as sawlog material in eastern white pine, northern white-cedar, and eastern hemlock ($p < .05$). The other minor softwood species were dropped from the model due to the lack of statistical significance when combined into a single species group ($p > .05$). Much like sawlog presence, there was a significant negative relationship between DBH and merchantable volume available as sawlog material for each species upon reaching a species-specific DBH ($p < .001$). The precision of the model was similar to the model developed for hardwoods, and it displayed a similar but larger negative bias (-0.0444). The RMSE for the model was 0.10 and the R^2 was 42%. When the models are combined using a probabilistic approach, the maximum ratios of sawlog volume to merchantable volume were reached between 41 cm and 46 cm DBH for the three species, but only Eastern white pine and Eastern hemlock demonstrated the presence of sawlog material beyond 60 cm, with the ratio dropping to 8% for northern white-cedar (Figure 1.8).

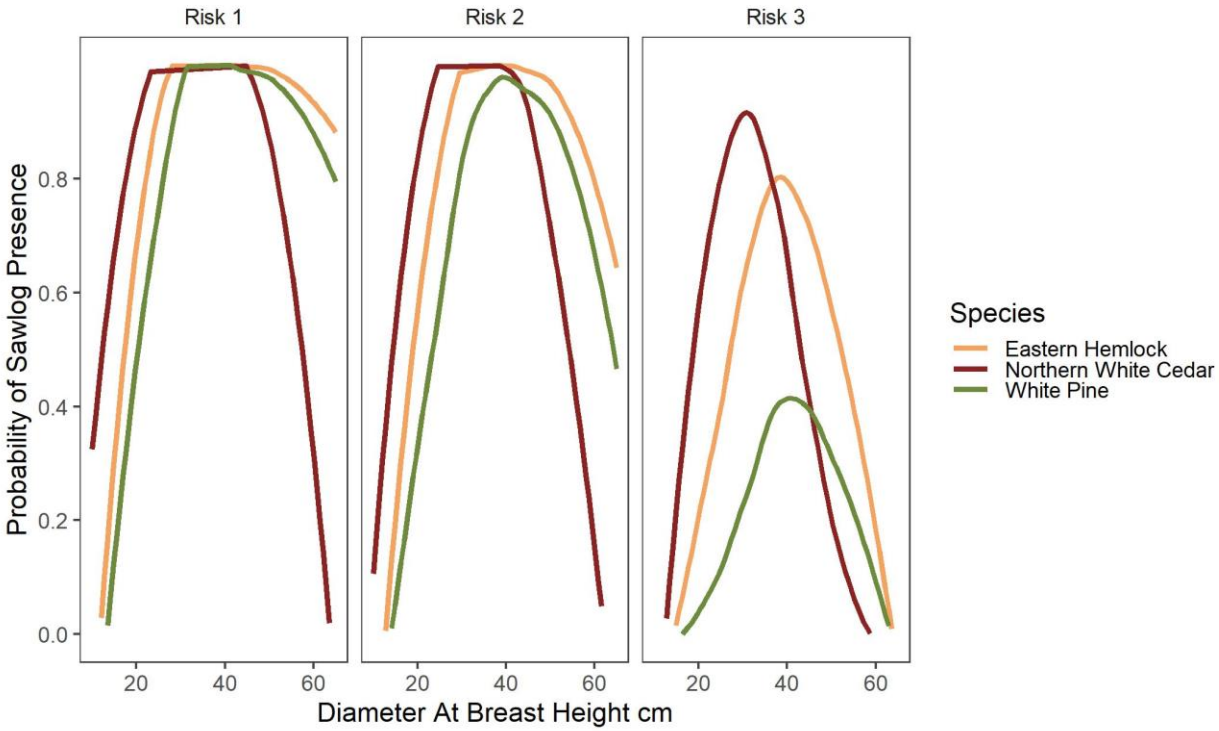


Figure 1.7 Effect of diameter on likelihood of sawlog presence for Eastern hemlock, Northern white-cedar, and Eastern white pine trees that demonstrate ideal stem form across across three risk classes with a mean height of 18 m and CSI of 18.

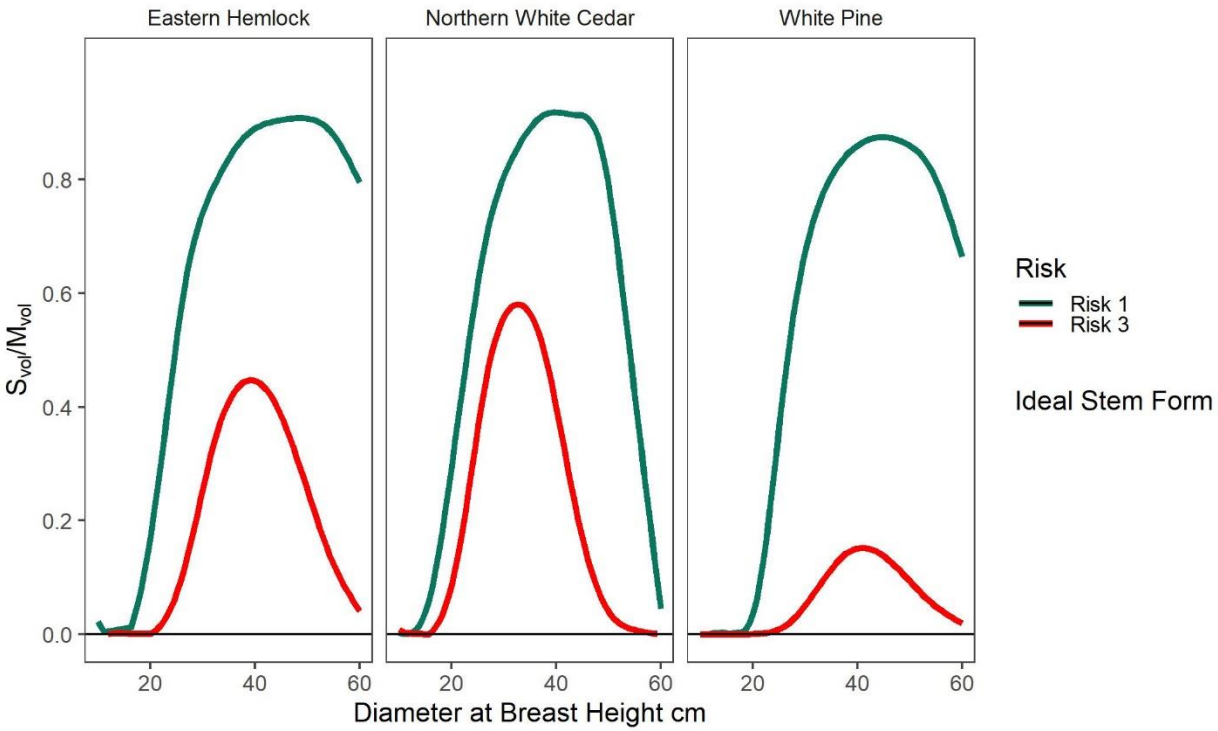


Figure 1.8 The percentage of merchantable volume composed of sawlog quality material for risk class one and risk class three eastern hemlock, northern white-cedar, and white pine stems demonstrating ideal form across diameter classes using a two-step probabilistic model that multiplies the predicted value of equations 15 and 18. Figure represents trees with a mean height of 18m and a CSI of 18.

Discussion:

In our sample we found a high degree of variability in stem forms that is consistent with previous observations made by Castle et al (2017). In the species examined in prior studies, stem form distribution patterns identified were similar to the ones found in this study. As an example, red maple demonstrated a propensity to display multiple stems (35% of sample) due to stump sprouting, a well-established method of regeneration for the species in managed stands (Keyser & Loftis, 2015; Keyser & Zarnoch, 2014). A large amount of stump sprouting was also found in red oak, consistent with oak species using stump sprouts as a common form of regeneration after harvests (Ward & Williams, 2018). Sugar maple and yellow birch both had a high presence of stems displaying ideal form, 71% and 49%, respectively, which is consistent with their reputation as high-value species managed for maximum financial return.

In species not included in the Castle et al. (2017) study, we found stem form distributions that are consistent with biological expectations. In the hardwood sample, approximately 70% of the American beech were high risk, which is consistent with the high prevalence of beech bark disease in the region (Kasson & Livingston, 2012; Morin et al., 2007). In the softwood sample, Eastern white pine stems that were not of ideal form most frequently demonstrated significant curves or sweep on two axes, which is consistent with the results of damage caused by white pine weevil that destroys the tree's leader (Verrez et al., 2010).

While model accuracy was not strong for predicting stem form and risk, the variables identified are consistent with expectations that site characteristics may play a role in stem form and risk. Growing degree days, frost free period, and mean growing season temperatures are all variables that have shown significance in other studies as variables that play a role in site quality (Hennigar et al., 2017). The variability in the sample and the necessity of combining large groups of species with different ideal growing conditions was a potential driving factor in the weak performances of these models.

Stem form and risk assessment proved to be a significant and effective predictor for sawlog presence and sawlog volume when paired with other tree-, site-, and stand-level characteristics for all tested hardwood species ($p < .001$). Models for predicting sawlog presence performed especially well with AUC values of .93 for hardwoods, .90 for spruce and fir, and .89 for the other softwood species tested when using only fixed effects. The model for predicting sawlog presence in hardwoods had a tendency towards errors of commission, but the model still accurately predicted sawlog presence in the sample data 86% of the time. The spruce and fir softwood presence model had a 90% accuracy rate with a tendency towards errors of omission. The model for other softwoods had an 84% accuracy rate with a tendency towards errors of omission. This strong performance highlights the importance of a two-stage approach and the usefulness of stem form as well as risk.

Consistent with the findings in Castle et. al (2017), all species examined demonstrated an increasing percentage of merchantable volume as sawlog material when diameter increased until a species-specific threshold was reached, at which point the percentage of predicted merchantable sawlog volume as sawlog material began to decrease at rates specific to each species. For example, Eastern white pine and Eastern hemlock tended to maintain a higher degree of sawlog material as a percentage of merchantable volume across larger diameters than Northern white cedar. This is likely a result of species-specific features of Northern white-cedar, where previous research has demonstrated that the probability of internal decay increases significantly with increases of diameter at breast height (Fraver et al., 2020). Previous studies have also shown Eastern hemlock and Eastern white pine to have lower probabilities of decay due to biological characteristics such as drought tolerance in the case of white pine and leaf longevity in the case of hemlock (Frank et al., 2018). The models developed for predicting sawlog volume in this analysis do not incorporate sawlog grades, which is a potential limitation when compared to other proposed methodologies (Bédard et al., 2018). However, sawlog

grades can vary dramatically for a given species, intended end use, and local markets, which may create higher variability in application in comparison to stem form and risk.

Using a much larger sample size across a greater number of species than in previous similar studies, form class became a less significant predictor of sawlog presence and the percentage of volume present as sawlog quality. Only the presence of ideal stem form became significant, as all other form classes displayed a high degree of variability in sawlog presence and percentage of volume of sawlog material. This effect was likely compounded by the fact that multiple risk classes proved to be significant, and there were potential redundancies between the risk and form class analyses originally proposed by the Northern Hardwood Research Institute that were used in this study. Due to the fact that the risk classes incorporated forking, external decay, mechanical damage, and other factors that would clearly affect a tree's merchantability, only the presence of a straight stem with no significant lean or sweep proved to be a significant predictor when observing form (Castle et al., 2017; Pelletier et al., 2016b).

All hardwood species were grouped together in this analysis for the prediction of form, risk, sawlog presence, and the percentage of merchantable product as sawlog quality. This led to a high degree of variability in the sample both within and among species. As a result, tree-specific metrics such as height and diameter at breast height as well as one-sided competition metrics such as the basal area of larger trees proved significant ($p < .05$), while additional site and climate variables were less effective predictors. Each species has different preferred site characteristics, varying flood and drought tolerances, and contrasting growth and regeneration strategies, which create highly distinct species-specific characteristics. In contrast, the spruce-fir samples showed much less variability, and this was reflected in form and risk models that demonstrated greater predictive power. Additionally, site-level variables such as mean annual precipitation and the number of growing degree days proved to be significant predictors ($p < .001$). This suggests that using a larger sample size might be necessary to create

more accurate species-specific models used to predict the form and risk class of individual stems and the percentage of merchantable volume composed of sawlog quality material.

The models developed in this analysis were relatively robust, especially in the prediction of sawlog presence. However, there was a high degree of variability in the sample with stem quality being an extraordinarily complex phenomenon that is dependent upon a stand's disturbance history, site quality, species composition, and likely tree genetics. This variability would likely remain or even increase as sample size increased, but the ability to detect the role of stand- and site-level variables would probably improve. Better predictions of product ratios may have the ability to inform management decisions and improve assessments of economic value. This may be particularly important in this region, as improved inventory data that explains more about the merchantable product available can lead to more efficient and effective tactical and long-term harvest planning (Borders et al., 2008; Holopainen et al., 2010). Additionally, the improvement of these models would allow for their incorporation into growth and yield models and the ability to predict changes in sawlog presence and volume over time as tree- and stand-level conditions change. For example, Castle et al. (2018) demonstrated the influence of stem form and risk on tree growth, which had important implications for projections of future stand-value in areas dominated by commercially important hardwood species. Castle et al. (2018) also demonstrated that the incorporation of form and risk affected predicted stand values in simulations, and this study highlights that form and risk observations can indeed be an effective tool for improving projected stand values.

There were some important limitations to this study due to the coarse nature of the climate and site quality data that was used. The use of multiple observers and the general difficulty of field merchandizing standing timber were additional confounding factors. We did not incorporate stand history into this study and all of the samples were from actively managed sites, so there is no certainty

of the generally representative nature of the sample across the region. Finally, there was no internal assessment of stem quality, so the estimates of sawlog volume might be optimistic.

In general, due to decreased variability in the sample, more robust models were developed for spruce and fir than those developed for hardwoods and other softwood species. While model performance for hardwoods improved when compared to the existing literature, a larger sample size and an ongoing monitoring program providing a more extensive database of observation would likely increase the ability to create more robust and species-specific models. Form and risk factors, along with tree-, stand-, and site-level characteristics, are clearly strong predictors of sawlog presence and the percentage of merchantable volume that is made up of sawlog quality material, but the quality of the models would be increased by future work that incorporates a larger and more diverse sample allowing for the development of species-specific models for hardwood and softwood species.

CHAPTER 2

INFLUENCE OF FIELD INVENTORY METHODS FOR ASSESSING STANDING SAWLOG VOLUME AND VALUE IN CONTRASTING MANAGED MIXEDWOOD STANDS OF NORTHERN MAINE

Abstract

The effectiveness and quality of sustainable forest management decisions is highly dependent upon the availability and nature of forest inventory data. Net present value calculations and harvest schedules are particularly reliant on representative and robust inventory data that accurately represents both the merchantable volume and product potential in forests under management. Past research has primarily focused on understanding the most effective inventory methods to estimate volume, species composition, and other common stand-level characteristics for use in both management planning and growth and yield modeling. However, to our knowledge, the effect of different field inventory methodologies commonly used in Maine on estimating product ratios has yet to be examined in detail. This is especially problematic in hardwood dominated forests due to the high degree of variability in both quality and value in individual trees. This high level of variability makes merchantable volume alone an insufficient predictor of sawlog quality material in hardwood species. This analysis seeks to examine how the inventory method used for field merchandizing influences estimates of stand value, sawlog volume, and potential product ratios in routine forest inventories. By using four unique and contrasting field merchandizing methods as well as a widely used regional growth model (e.g. Forest Vegetation Simulator), we observed high levels of variability in estimates of sawlog volume and stand values across ten hardwood-dominated forest stands in northern Maine. The estimated marginal mean sawlog volumes ranged from 39.27 m^3ha^{-1} to 110.45 m^3ha^{-1} across methods. The estimated marginal mean value across the four field methodologies and the FVS outputs ranged from \$2,582 ha^{-1} in the most conservative method to \$5,596 ha^{-1} in the least conservative. This demonstrated how the specific methodology chosen affects estimates of net present value, harvest planning when sawlog volume and

net revenue are used either as constraints or for optimization in modeling, and inventory planning when a cost-plus-loss analysis is used for investing resources into field inventories. Given the reality of fixed operating costs, constantly changing product specifications at mills, and the instability of the value of timber as a commodity, forest managers need consistent and reasonable estimates of sawlog volume as well as standing value in order to achieve their economic and ecological goals. The observed differences in variability and efficiency across the four field merchandizing methods and the regional growth model FVS show the real need to better understand the costs and benefits of different inventory methods as well as the need for further research on the accuracy and practicality of methods used in the field today. Of the methodologies tested, using stem form and risk to model product ratios is recommended due to reasonable estimates of sawlog potential, a framework that is easy to integrate into existing inventories, and a tool that may be used to predict sawlog potential in future stand conditions.

Introduction

The effectiveness and quality of sustainable forest management decisions is highly dependent upon the quality of forest inventory data. This data is the most valuable when estimates of product potential are included. Accurately predicting yield values for both long- and short-term operational planning depends on a reasonable estimate of stand values and potential product ratios (Holopainen et al., 2010). Harvest scheduling is also dependent upon inventory data that accurately represents both the merchantable volume and product potential in forests under management. When the goal of harvest scheduling is maintaining an even flow of sawlog volume or net revenue, forest managers need reasonable estimates of those values (Bettinger, 2009).

Existing research has demonstrated that high error rates in inventory data can lead to meaningful losses in value due to suboptimal decision making (Duvemo & Lämås, 2006). Scheduling stand improvement and other silvicultural treatments in hardwood stands requires a high degree of confidence in the quality of the wood in a stand. This is especially difficult on a landscape scale when

there are either high error rates in estimates of product potential or no estimates of product potential at all. Coarse data can lead to less informed decisions, which can directly lead to decreased profitability in operations (Borders et al., 2008).

Forests are complex and highly variable, particularly naturally regenerated stands that are often mixed species as well as multi-cohort. Error is a known byproduct of any inventory in these forest types. In order to achieve an even flow of sawlogs and revenue in management planning, a planner must have reasonable estimates of the error rates in the inventory data (Eyvindson & Cheng, 2016). Planners need to be able to report these statistics to other stakeholders and to understand how accurately they are able to compare alternative management decisions. Previous analyses have demonstrated how it is possible to control for error with high quality inventory data in a single-species plantation setting (Borders et al., 2008), but to our knowledge this has not been successfully demonstrated in highly variable multi-cohort, managed stands with a complex species composition primarily dominated by hardwood species.

Increased uncertainty makes yield value estimates less accurate and can lead to poor management choices. Measurement error in field inventory data has been shown to be a large source of uncertainty in estimates of net present value for even-aged stands (Holopainen, 2010a). Existing research has also demonstrated that errors in inventory data lead to more bias than either fluctuating stumpage prices or yield modeling estimates. The bias in net present value calculations due to errors in inventory data in multi-cohort, managed stands with complex species composition has not been tested. It is reasonable, however, to assume that such errors would lead to bias in these stands as it does in stands with simpler structure.

Field inventory methods are not chosen solely on accuracy but also on cost and efficiency. This is often done in a cost-plus-loss analysis. In a cost-plus-loss analysis, the cost of decreasing error rates in inventory data is balanced against the losses incurred from operational and long-term planning with

increased uncertainty (Holopainen et al., 2006). This is often unfeasible in forests where unknown variability makes error rates difficult to assess, especially when institutional resources may limit the available options.

In practice, field inventories often do not include any specific method for estimating product ratios. In these instances, merchantable volume alone is used to approximate product potential (Fortin et al., 2009). The standard outputs of regional growth and yield models like the Forest Vegetation Simulator (FVS) are frequently used to drive decisions concerning stand operability without any field measurements of sawlog volume (K. George et al., 2022). Field inventory methods are often chosen based upon institutional preference without sufficient regard paid to the variability and accuracy of the methodology (Rice et al., 2014).

Merchantable volume alone is often not an accurate predictor of value in hardwood stands. There are high degrees of variability in value and sawlog potential among sample plot locations and the stems within each sample plot (Castle et al., 2017). Sawlog presence and quality are not uniform across stands of equal volume, so value estimates will vary accordingly. This is not accounted for in regional growth and yield models like FVS. Additionally, estimating error rates in sawlog volume predictions requires that some method of field merchandizing be incorporated into field inventories. This is of particular importance in the eastern North American hardwood forest due to high variability that is a product of diverse species composition, past partial harvesting, and the reliance on natural regeneration (Olson & Wagner, 2010).

In addition, hardwood species are expected to increase in presence throughout eastern North America due to the influence of climate change and the presence of certain species like American beech (*Fagus grandifolia*) that have already shown greater abundance across the region (Bose et al. 2017). Current harvesting practices will have to be adapted to accommodate the shifting of species compositions toward greater hardwood presence in Maine forests. This change in composition had been

driven by both natural disturbance events such as spruce budworm outbreaks and disturbance regimes associated with harvesting that are inconsistent with the natural low levels of disturbance over long intervals found in unmanaged forests. Approximately 59% of merchantable carbon stock in Maine were hardwoods in 2017, which is an increase from 40% in 1975, demonstrating the increasing economic importance of their proper management (Chen et al., 2019). Previous studies have predicted that as much as 68% of the merchantable carbon stock in Maine may be composed of hardwoods by 2057 (Chen et al., 2019).

Hardwood species are capable of producing a large range of products with different market values, from pulp to veneer (Luppold et al., 2018; Luppold et al., 2004). Between 1996 and 1999, approximately 1.6 million m^3 of hardwood sawtimber and 4.7 million m^3 of hardwood pulpwood was harvested in Maine annually (W. G. Luppold & Sendak, 2004). In 2011, the North East State Foresters Association reported that approximately 1.2 million m^3 of hardwood sawtimber and 6.6 million m^3 of hardwood pulpwood was harvested in Maine (*The Economic Importance Of Maine's Forest-Based Economy 2013*, 2013). In these reports sawlogs made up approximately 16-25% of harvested volume. The differences may be driven by markets or methodologies, but both demonstrate that sawtimber is a minority of volume harvested.

The value of pulp and sawlog products varies widely. Mill rates in western Maine in September of 2021 for sugar maple (*Acer saccharum*) veneer were as high as \$1,100/m^3, while a similar volume of pulp could be worth as little as \$38/m^3. Sugar maple sawlogs that are not of veneer quality ranged in price from \$170/m^3 to \$425/m^3 in the same region depending upon log grade (*Log Yard Prices*, 2021). Similar patterns are seen with other commercial hardwoods in the region such as red maple (*Acer rubrum*), yellow birch (*Betula alleghaniensis*), and white ash (*Fraxinus americana*).

The proportion of merchantable volume in a tree that can be sawn into lumber is highly variable, and commonly, a tree may have no sawlog potential due to poor form or specific defects.

External and internal stem defects are difficult to predict and vary widely between individual trees. Even in observed standing trees, it can be difficult to assess internal defect and decay prior to felling (Frank et al., 2018). The situation is further complicated by the wide variety of stem forms commonly exhibited in hardwoods including forking, lean, sweep, and stump sprouting (Castle et al., 2017).

Methods used for silvicultural assessments of tree vigor alone have been shown to be poor predictors of the standing value of timber (Havreljuk et al., 2014). Metrics used for identifying trees for the purpose of stand health and making stand-level silvicultural decisions do not necessarily correlate strongly with sawlog potential. Trees of high vigor may have external defects that prevent them from producing sawlogs, while low vigor trees near their natural rotation age may produce excellent sawlogs if extracted prior to mortality.

Past research has primarily focused on understanding the most effective inventory methods to estimate volume, species composition and other stand-level characteristics for use in management planning as well as growth and yield modeling. An extensive body of literature exists on sampling methods including variable-radius sampling, fixed-area plots, and 3P sampling (Kershaw et al., 2017; Vries, 1986). Forest stratification has been covered in detail with work focusing on ways to reduce sampling error (Iles, 2003; A. Robinson & Hamann, 2011). The calculation of individual tree estimates of volume, stem taper, crown width, and a variety of other features has also been widely studied. Finally, a variety of stand density indices and other measures of competition have been published (Aaron R. Weiskittel et al., 2011; Kershaw et al., 2017; Woodall et al., 2005). To our knowledge, while estimates of total merchantable volume have been studied, the effect different field inventory methodologies have on estimating sawlog potential has not been examined in detail. This analysis seeks to examine how the four inventory methods used for field merchandizing influence estimates of stand value and sawlog volume in routine forest inventories. Variance of estimates at the plot level and the driving factors behind the different mean estimates utilizing each method will be analyzed. The relative efficiency of

each method will also be examined to understand key tradeoffs between effort and variability across methodologies.

Field Merchandizing Methodologies

Currently in Maine there are several methodologies being used by different institutions to estimate product ratios based on past preferences and available resources. Common field merchandizing methods include (1) acceptable sawlog grade (ASG) where a stem is designated either sawlog grade or non-sawlog grade, (2) stick cruising where trees are visually assessed in 2.44 m (8 ft) increments and (3) merchantable height determination where the stem is scaled, and the observed sawlog portion is measured with a hypsometer. These methodologies are used to estimate the standing sawlog volume of forests during both landscape-scale inventories and pre-harvest stand assessments. The literature that examines field merchandizing methodologies in use for predicting sawlog volumes or product potential is scarce, particularly in hardwood dominant stands found in eastern North America.

Form and risk classification is a methodology that has been proposed and studied and is being measured by several large landowners in Maine. The ability to make sawlog volume and stumpage value estimates based on this data has previously been limited due to the fact that predictive models have only been developed for four common hardwood species and have not been applied to any softwoods (Castle et al., 2017; Pelletier et al., 2016a). Models using form and risk class for 14 commercially important hardwood and softwood species in Maine were developed in chapter 1, allowing form and risk observations to be used as a field merchandizing methodology.

When no field merchandizing is conducted, sawlog and pulp estimates provided by regional growth models such as the Forest Vegetation Simulator (FVS) are often used. FVS is the growth and yield modeling platform maintained by the United States Forest Service. The platform is one of the approved growth and yield programs approved for a variety of official uses including growth projections for

carbon offsets in the State of California (*U.S. Forest Protocol Approved Growth and Yield Models | California Air Resources Board*, 2022). To our knowledge, the sawlog volume estimates provided by regional models like FVS have not been directly compared to field measurements across a range of stand conditions.

To address this issue, this analysis evaluates plot-level estimates of sawlog volume and mean stumpage values using Maine statewide averages for sawlogs and employing the four identified contrasting field inventory methods used in Maine. These values are then compared against standard outputs from the widely used regional forest growth and yield model (i.e., the Northeastern Variant of the Forest Vegetation Simulator; FVS-NE) (Crookston & Dixon, 2005). This data is then expanded to the stand level to examine the variability in total sawlog volume estimates when compared to estimates of total merchantable volume in forest stands. Once key drivers of sawlog volume and stumpage value are evaluated and the contrasting methods assessed, relative efficiency between methods will be analyzed to understand key benefits and tradeoffs among the different methodologies.

Methods:

Study Area

The study was conducted in Maine and consisted of 10 research stands in two distinctly different geographic regions of the state (Figure 2.1). The primary species composition of the Acadian Region includes an abundance of softwood species, including red spruce (*Picea rubens*) and balsam fir (*Abies balsamea*), as well as hardwood species, including shade tolerant species such as red maple, sugar maple, American beech, and yellow birch. In addition, intolerant species such as paper birch (*Betula papyrifera)*, grey birch (*Betula populifolia),* and quaking aspen (*Populus tremuloides*) (Seymour, 1995) are also common. Eastern white pine (*Pinus strobus)*, Eastern hemlock (*Tsuga canadensis*), and Northern red oak (*Quercus rubra)* are primarily found in the southwestern portion of the region (Eyre, 1980).

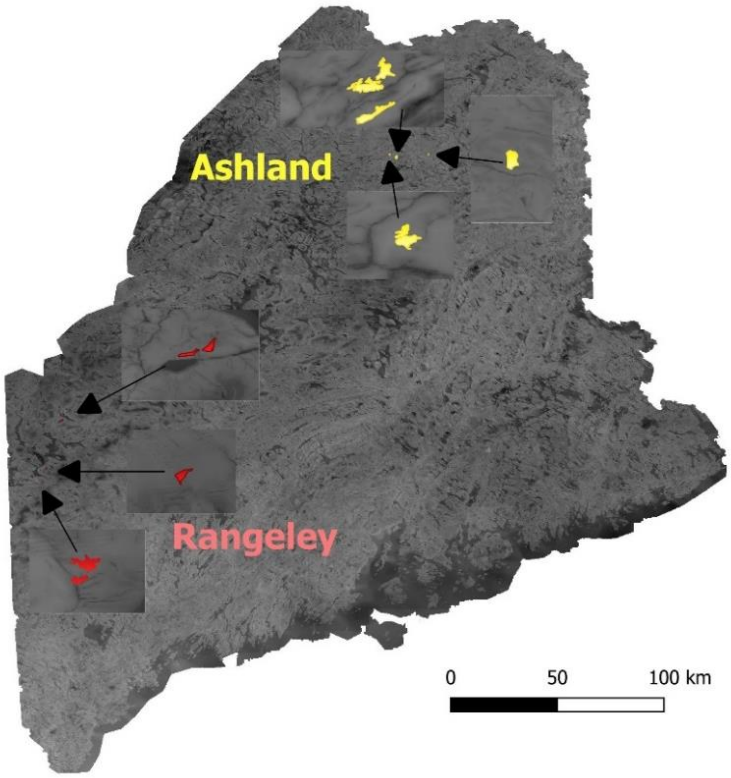


Figure 2.1 Locations of the ten hardwood dominate stands identifies by region against the biomass growth index. Arrows point to actual stand locations.

In Maine, soils are generally classified as Spodosols and tree species in the region are distributed due to a number of factors with soil drainage being a key component (Fisher & Binkley, 2000). Podzol

soils result from heavy rainfall and cool temperatures and can vary from acidic sandy loam to shallow and stony soils (Loo & Ives, 2003). Annual precipitation can vary widely across time and regions within the state with annual precipitation ranging from 60 to 160 cm per year over the past two decades (Birkel, 2021). The region's topography is varied, with elevations ranging from sea level to 1600 m.

In most of the region's forests past harvesting of varying intensity is common. All of the ten stands in the study area have been actively managed to favor commercial species. In the Ashland region, stand 1A had a diameter limit harvest in 1986 on the southern half of the stand and the northern side was clear cut the same year. Stands 2A, 3A, and 5A had previous entries between 1999 and 2001 that were either a single tree selection harvest or a salvage harvest after a blowdown event. Stand 4A received a crown thinning in 1999. In the Rangeley region, stands 1R and 2R had group selection harvests in 1996. Stand 3R had a group selection harvest in 2021, immediately prior to conducting the inventory. Stands 4R and 5R had previous stand improvement harvests in 2004 and 2010.

Merchandizing methodologies

The four merchandizing methodologies were chosen based on their current or potential application for use in Maine (Table 2.1). In this region, there are three commonly used methods for predicting the volume of sawlog material in a hardwood stand. The first is the Acceptable Sawlog Grade system (ASG), which is often referred to as acceptable/unacceptable growing stock (AGS/UGS). The name Acceptable Sawlog Grade was used in this analysis to distinguish the act of predicting merchantable value from the act of marking trees specifically for silvicultural purposes where variables beyond merchantable value are often of concern (*Ontario Tree Marking Guide*, 2004). In practice, trees are marked as either sawlog grade or unacceptable based on whether at least one 2.43 m (8-ft) sawlog is present upon visual observation. The entire merchantable volume of the stem is then assumed to be of that specified quality.

Table 2.1 Description and examination of potential advantages and disadvantages for tested field merchandizing methods.

Method	Description	Potential Advantages	Potential Disadvantages
Acceptable Sawlog Grade (ASG)	The tree is inspected to see if it has the potential to create any sawlogs. If the tree is observed to have at least one 2.44 m (8-ft) section of sawlog quality on the stem it is considered ASG.	The entire length of the stem does not need to be assessed once one 2.44m sawlog is found in the stem.	No distinction is made about how much sawlog material is present in the tree. Thus, the entire portion of the stem that is of merchantable diameter is considered to have sawlog potential.
Stick Cruising	Every 2.43 m (8-ft) section of the tree stem is categorized as either sawlog, pulp, or cull. The heights and lengths of sections and the minimum upper stem diameter of each section are visually estimated.	The portion of the stem that is sawlog quality can be estimated in 8-foot sections, leading to a more accurate representation of the quality of wood throughout all portions of the stem.	All measurements and product calls are visually estimated, which can lead to significant error in the difficulty of product calls, the number of logs predicted, and the location of the minimum upper stem diameter (Westfall et al., 2016). An additional source of error is the skill of the forester subjectively predicting whether a portion of the stem is sawlog quality.
Merchantable Stopper Heights	The exact length from the beginning to the end of potential sawlog material in the main stem is measured for each tree using a digital hypsometer.	The potential sawlog portion of the stem is measured with the exact starting and end locations of the sawlog portion being noted, allowing for a more precise measurement of sawlog volume.	The source of potential error is primarily with the skill of the forester subjectively predicting whether a portion of the stem is sawlog quality. As with stick cruising, there is a high level of subjectivity in product predictions.
Form and Risk	The stem form and risk of each tree is recorded based on a field guide which clearly articulates the definitions for each category using a decision tree along with images.	The easy-to-use field guide takes subjectivity out of the process. Cruisers are not required to have any log scaling knowledge.	The largest source of potential error rests in the accuracy and representative nature of the models utilized to predict sawlog volume using the form and risk observations.

The second system is stick cruising, in which trees are visually scaled in 2.44 m (8-ft) sections. The third method is merchantable height, in which the length of the sawlog portion of a stem is measured with a digital hypsometer (Table 2.1).

The fourth field method used in this study relied on the use of stem form and risk classification. The Northern Hardwood Research Institute developed a detailed classification system that organized hardwood trees into classes based on stem form and mortality risk (Pelletier et al., 2016a) (Table 2.2, Figure 2.2). In Maine, this classification system was used to develop coefficients for predicting sawlog potential for Northern red oak, yellow birch, red maple, and sugar maple (Castle et al., 2017). This approach offers a new way to incorporate sawlog volume predictions into forest inventories in a simple-to-understand and easily implemented system with less subjectivity and possibly reduced error when compared to existing methodologies.

While form and risk data has been collected by several land managers in Maine, sufficient assessments have not previously been conducted to be able to incorporate this information into sawlog estimates. This method was extended to additional species and refined in Chapter 1 in order to include sawlog volume and value estimates using stem form and risk observations. See Appendix C for more detailed field protocols.

Table 2.2 NHRI form classes (Castle et. al, 2017)

Classification	Description	Classification	Description
Stem form classification			
F1 	Ideal form; single straight stem; no significant fork on the first 5 m of the stem	F5 	Poor form; multiple stems are present between 0.3 and 1.3 m of the stem
F2 	Acceptable form; sweep on two axes or one significant curve on the stem	F6 	Poor form; single stem on the first 5 m; significant lean >15° on the vertical axis
F3 	Poor form; large branches on the first 5 m; large branches may have roundwood products	F7 	Acceptable form; significant fork between 2.5 and 5 m of the stem; large branches may have roundwood products
F4 	Unacceptable form; multiple stems or branches on the first 5 m; large branches do not have roundwood products	F8 	Poor form. Multiple stems present on the first 0.3 m of the stem
Stem risk classification			
R1 — low risk	Potential defects: crook, sweep, white face scar, burl	R3 — high risk	Potential defects: same as R1 and R2, cracks, open wounds, fruiting bodies, reduced crown ratio, dead branches
R2 — moderate risk	Potential defects: same as R1, small dark face scar, fire scar, animal damage, suppression	R4 — very high risk	Potential defects: same as R1, R2, and R3, multiple severe wounds (those included in R3)

Figure 3. Determination key for determining risk of losing vigor ("R" rating)

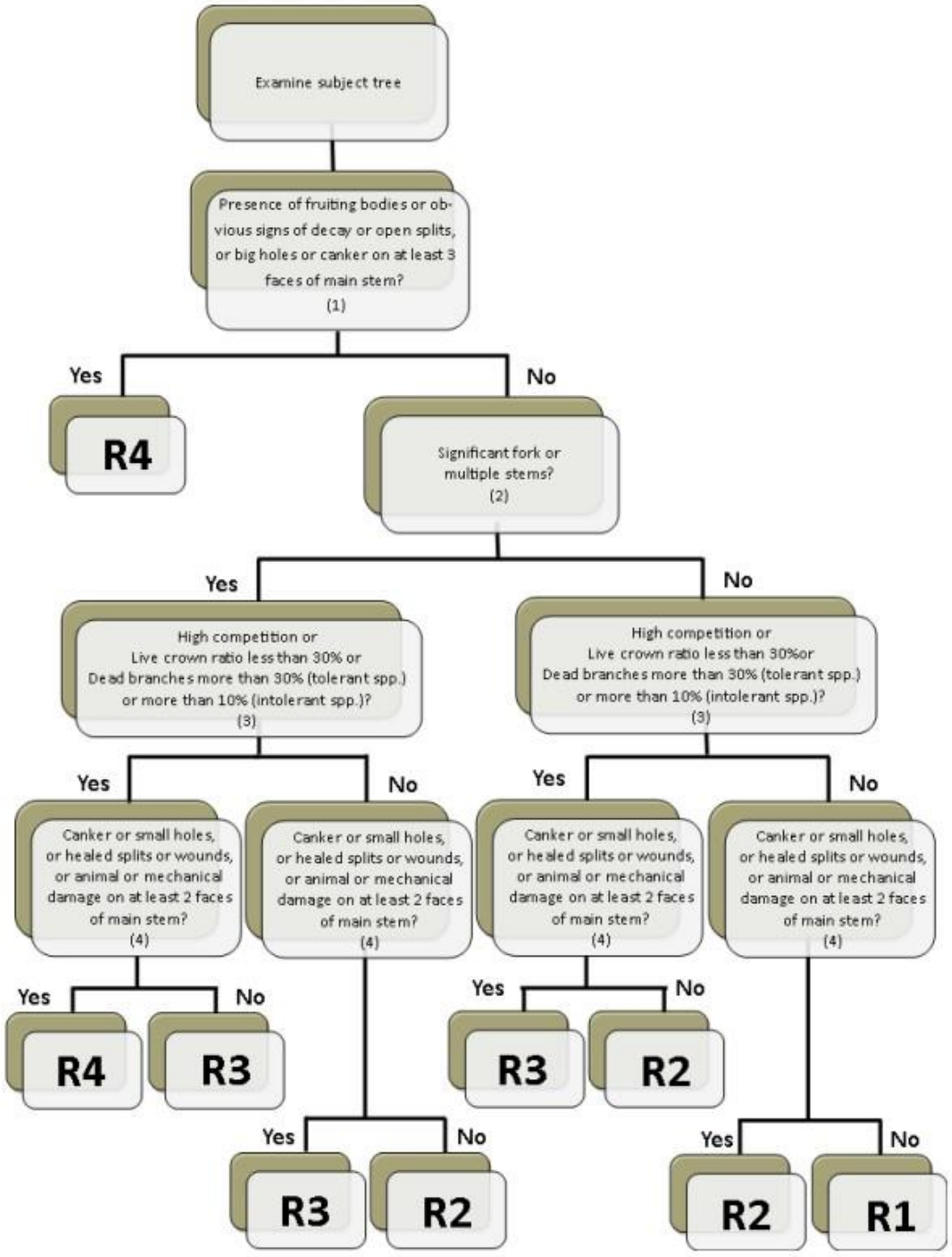


Figure 2.2 NHRI Risk Class Decision Tree (Pelletier et al., 2016b).

Field Measurements

In each study area, plot locations were assigned by placing points in a 100 m grid beginning at a random starting point. This method was chosen because of its common use for forest inventories in industry (Vries, 1986). The study included 120 plots with a sampling intensity that averaged one plot per 1.04 ha. At each plot location, variable radius sampling was used, and a 4.6 metric basal area factor (BAF) prism (20 BAF in English) was used to count "in trees" and then a second sweep was done with a 18.4 metric BAF prism (80 BAF in English) to capture a subsample of heights. All "in" trees greater than 10 cm had their diameter at breast height (DBH) measured and their species recorded. All additional "in" trees for the 18.4 BAF prism had the total tree height measured using a digital hypsometer (Haglof Vertex 3). If a tree forked below breast height, each stem after the fork was recorded as a unique tree. The walk-through method was used for any plots near the stand boundary (Ducey et al., 2004).

After this initial data was collected, each field merchandising method was used at the plot in a randomized order (Appendix C). From plot center and with a timer, every tree was merchandized using the first randomly selected method. This procedure was then repeated for the following three methods. Randomization of the methods for each plot was done using a random number generator prior to entering the field to reduce bias. In instances where the beginning and end of the merchantable section of the tree was measured with a hypsometer (merchantable stopper heights), the stopper heights were not revealed until after all four methods were completed, while an independent yet consistent observer was used for stick cruising given its dependence on visual estimation. This was done to ensure that the individual estimating the number of logs visually was not biased by the measured merchantable heights.

For trees with no measured heights, a species- and plot-specific modified mixed-effects model was utilized, similar to that used in a prior study (A. P. Robinson & Wykoff, 2004) (see Appendix B). For spruce-fir trees, no merchantable height measurements or log estimates were made in the field unless the tree was determined to be sawlog quality only to a specific location that was clearly below where

the tree reached its minimum sawlog diameter. This was done because estimating the location of the minimum diameter of the tree has previously been shown to be highly inaccurate (Westfall et al., 2016), and most of the observed spruce-fir samples were either cull, pulp, or sawlog quality for the entire length of the stem. Regional taper equations were used to estimate the height at which the minimum sawlog diameter was reached on trees determined to be sawlog quality (Appendix C) (Weiskittel & Li, 2012; Kozak, 2004; Li et al., 2012).

Merchandizing

Board foot estimates were obtained by merchandizing each tree into 2.43 m (8-ft) logs and applying the International ¼ rule (Appendix B), which was utilized due to its common commercial usage in Maine. If two 2.43 m logs could not be created, the remainder of the sawlog portion of the stem was added to the log with the smallest end diameter for all methods. When field observations included portions of the stem above the location of the minimum merchantable sawlog diameter, sawlog potential was reduced to the estimated height of the minimum estimated merchantable sawlog diameter using the regional taper equations. For stick cruising measurements, when a log did not meet diameter requirements it was reduced to pulp. For form and risk analysis, the percentage of sawlog content was calculated based on previously developed equations (Chapter 1).

For all methods, if a minimum log length of 2.4 m was not met for a tree the entire volume was reduced to pulp. The 2.4 m measurement was chosen to standardize estimations of board feet across methodologies, allowing for consistent comparisons. Because stick cruising requires calculations be made in 2.4 m segments, not reducing the tree into 2.4 m sections for the additional methods would have negatively biased their value estimates because the International ¼ log rule potentially biases longer logs even if they contain the same volume as two shorter logs. While modern mill requirements often require logs of varying lengths beyond 3 m, 2.4 m is used as the standard in most stick-cruising

operations in Maine and thus was assumed for all methods examined in this analysis. All merchandizing functions were written in the R programming language (R Core Team, 2021).

For the FVS estimates of sawlog board feet and pulp volume, minimum merchantable diameter specifications were adjusted for each species to match the diameters used for the other methods. The volume equations used by FVS are from the National Volume Estimator Library (*National Volume Estimator Library (NVEL)*, 2021).

For stem form and risk predictions, two contrasting approaches were examined. The first is a probabilistic approach where the percentage of sawlog volume is estimated to be $S_{vol}/M_{vol}^{y} = Pr(Sawlogs) \times S_{vol}/M_{vol}^{x}$ based on species-specific equations developed in chapter 1. S_{vol}/M_{vol} represents the proportion of merchantable volume that is predicted to be sawlog quality. Pr(Sawlogs) represents the probability of a tree having a sawlog.

The second model relies on the use of a two-stage binary approach. In the first stage the Pr(Sawlogs) model is used to determine the presence of a sawlog in a tree, but it is treated as a binary event. Sawlog presence is true when a species-specific threshold value is met. The threshold value was optimized to balance errors of omission and commission using the OptimalCutpoints package in the R programming language (Lopez-Raton & Rodriguez-Alvarez, 2021; R Core Team, 2021). In the second stage the species-specific S_{vol}/M_{vol}^{x} equation is applied to trees with predicted positive sawlog occurrence to determine the percentage of merchantable volume made up of sawlog material. All trees with a Pr(Sawlogs) value below the threshold value are considered to have a S_{vol}/M_{vol}^{y} value of 0. All Pr(Sawlogs) and S_{vol}/M_{vol}^{x} models used can be found in Chapter 1 and model tables containing coefficient values, standard errors, and goodness of fit statistics can be found in Appendix A.

The financial value of each tree was calculated by using reported statewide average prices for sawlogs and pulp in 2019 by species for the state of Maine (*2019 Stumpage Prices By Maine County/Unit*, 2021). Pulp values were determined by subtracting the estimated saw volume from the

estimated total merchantable volume for each non-cull tree and converting this volume to weight using the tree species' estimated green weight. The tree-level measurements were scaled up by multiplying the tree's predicted value by its expansion factor to obtain plot- and stand-level estimates of value and sawlog volume.

Statistical Analysis of Financial Value and Sawlog Volume

Once plot-level estimates of value were calculated for each merchandizing method, covariates were tested for statistical significance ($p < .05$) to ensure mean estimates of value were not being confounded by factors other than merchandizing methods in a way that would bias the estimated means. Basal area (m^2) ha^{-1} (BAPH) and the percentage of the stand made up of hardwood species (HW%) proved to be significant covariates. Since volume is calculated differently in FVS than the merchandizers built for the other four methods, volume (m^3) ha^{-1} was excluded as a potential covariate for this portion of the analysis.

A mixed model was used to account for the fixed effect of each merchandizing method and the random effects associated with variability among stands and plots within stands. Models were tested using all significant variables and interaction terms ($p < .05$). Variable inflation factors were tested. When only minor differences in model performance were identified, the simpler model was selected. Once all models were fit, a mixed-effects ANCOVA was conducted to assess differences in group estimated marginal means among merchandizing methods for both sawlog volume and stand value. Estimated marginal means (least square means) for this analysis were calculated using the emmeans package in R (Lenth, 2022) . Multiple comparisons were performed between methods using pairwise Tukey honestly significant difference tests (Tukey HSD). Differences were considered significant when $p < 0.05$.

Sawlog Volume

For sawlog volume the following linear mixed model was selected to estimate the estimated marginal mean plot-level sawlog volume for each field methodology and FVS:

$$[19] \qquad S_{vol} = \beta_0 + \beta_1 Method_6 + \beta_2 BAPH + \beta_3 Method * HW\%$$

where S_{vol} is the estimated plot level sawlog volume, the effect of *BAPH* is consistent across methods, and there is an interaction between each *method* and *HW%* (Table 2.3). Stand and plot were used as nested hierarchical random effects. The *HW%* interaction was included not only because the slopes of the field merchandizing methods differed, but also because FVS showed a completely different behavior than the field merchandizing methods. While sawlog volumes decreased as HW% increased for all field methods, there was a slight increase in sawlog volume as HW% increased in estimates provided by default FVS outputs.

Merchantable Value

The following linear mixed model was selected to estimate the estimated marginal mean plot-level value for each field methodology and FVS:

$$[20] \qquad Value = \beta_{10} + \beta_{11} Method_6 + \beta_{12} BAPH$$

where *Value* is the estimated value in \$/ha^{-1} and the fixed effect of *BAPH* is consistent across methods (Table 2.3). Plot within site were used as nested hierarchical random effects.

Efficiency

The efficiency of each inventory method was also evaluated. All methods were tested for efficiency using a raw efficiency score where:

$$[21] \qquad Efficency = Time_{Method} * SE_{Method}$$

where $Time_{Method}$ is the average seconds required to complete a method at the plot level and SE_{Method} is the stand-level standard error based on the plot-level observations. An ANOVA was conducted on mean efficiency for the contrasting inventory methods after accounting for region (western vs. northern Maine). Lower mean efficiency scores indicate higher efficiency.

Table 2.3 Parameter estimates for equations 19 and 20.

	Eq. 19 Sawlog Volume (m^3/ha^{-1})				Eq. 20 Stand Value ($ ha^{-1})			
Ind. Variable	Par.	Estimate	SE	t-value	Par.	Estimate	SE	t-value
Intercept	$\beta 0$	53.1434	15.1843	3.500	$\beta 10$	-91.17	608.2	-0.283
F/R Binary	$\beta 1$	0.0021	11.6980	0.000	$\beta 11$	-166.24	141.98	-1.171
F/R Prob	$\beta 1$	-1.2649	11.6980	-0.108	$\beta 11$	-285.26	141.98	-2.009
FVS	$\beta 1$	-41.3098	11.6980	-3.531	$\beta 11$	2002.36	141.98	14.103
Merch	$\beta 1$	-7.2159	11.6980	-0.617	$\beta 11$	-876.22	141.98	-6.171
Stick	$\beta 1$	-12.1371	11.6980	-1.038	$\beta 11$	-1051.60	141.98	-7.407
BAPH	$\beta 2$	3.1940	0.29201	10.938	$\beta 12$	156.85	141.98	13.934
ASG x HW%	$\beta 3$	-75.8201	15.8078	-4.796				
F/R Bin x HW%	$\beta 3$	-82.1997	15.8078	-5.200				
F/R Pr x HW%	$\beta 3$	-84.7555	15.8078	-5.362				
FVS x HW%	$\beta 3$	27.5880	15.8078	1.745				
Merch x HW%	$\beta 3$	-91.4580	15.8078	-5.786				
Stick x HW%	$\beta 3$	-91.7230	15.8078	-5.802				
Fit Statistics								
R^2		.84				.82		
RMSE		21.82				1003.36		
MAE		15.99				708.90		

***Note:** Ind. variable, independent variable; Par., parameter; F/R Binary, binary form/risk model; F/R Prob, probabilistic form/risk model; Merch, merchantable height; Stick, stick cruising; BAPH, basal area per hectare m^2; HW%, percent of volume composed of hardwoods (0-1.0); RMSE, root mean square error; MAE, mean absolute error. Acceptable sawlog grade was used as the reference level with which other methods were compared.

Results:

The 10 stands examined in this analysis varied in overall basal area and species composition. The mean values of BAPH and percentage of volume composed of hardwoods were 23.32 ± 1.84 m^2 and 83.5 ± 2.9%, respectively (Table 2.4). In the 118 plots merchantable volume had a mean value of 190 ± 8.9 m^3. The mean sawlog volume and financial values when averaged across all field methods and the FVS values were 190 ± 8.9 m^3 and $3603 ± 95, respectively (Table 2.4). Sawlogs comprised 32 ± 1% of the total

merchantable volume when averaged across all methods. The ranges and distribution of sawlog volume and stumpage value estimates varied between methods.

Table 2.4 Summary of study area stand and plot Level attributes.
BAPH, Basal area per hectare; QMD, Quadratic mean diameter; Merch vol, total merchantable volume; Saw_{vol}, total sawlog volume; S_{vol}/M_{vol}, proportion of merchantable volume that is sawlog quality.

Attributes	Values
Stands (n = 10)	
BAPH m^2/ha^{-1} (Mean; SD)[min, max]	(23.32, 9.17) [15.09, 35.20]
QMD cm (Mean; SD)[min, max]	(29.1, 8.5) [22.6, 34.18]
Percent HW (Mean; SD)[min, max]	(.83, .21) [.72, 1.0]
Relative Density (Mean; SD)[min, max]	(.41, .16) [.25, .62]
Trees ha^{-1} (Mean; SD)[min, max]	(444, 55) [256, 677]
Elevation m (Mean; SD)[min, max]	(487, 108) [357,622]
Plots (n = 118)	
BAPH m^2/ha^{-1} (Mean; SD) [min, max]	(23.62, 10.71) [4.59, 55.10]
QMD cm (Mean; SD) [min, max]	(29.8, 9.8) [11.4, 80.3]
Percent HW (Mean; SD) [min, max]	(0.85, 0.24) [0.0, 1.0]
Relative Density (Mean; SD)[min, max]	(0.41, .18) [.08, .96]
Trees ha^{-1} (Mean; SD) [min, max]	(432, 276) [9, 1183]
Merch Vol m^3 (Mean; SD) [min, max]	(190.51 96.45) [0, 518.5]
***Saw_{vol}** m^3 (Mean; SD) [min, max]	(61.9, 59.6) [0, 445.6]
***S_{vol}/M_{vol}** % (Mean; SD) [min, max]	(32, 25) [0, 99]
***Value** $/$ha^{-1}$ (Mean; SD) [min, max]	(3603, 2554) [0, 17181]

* Values averaged across all methods

Estimates of Sawlog Volume

In the model used to predict sawlog volume both BAPH and the interaction between HW% and methodology used were shown to be significant covariates. The slopes for each of the field merchandizing methods tested showed that there was a negative correlation between sawlog volume estimates and HW% at the plot level. The model showed a positive correlation between sawlog volume and HW% for estimates provided by FVS. The predicted differences among estimates for the tested field merchandizing methods decrease as the percentage of volume composed of hardwoods decreases (Figure 2.3*).*

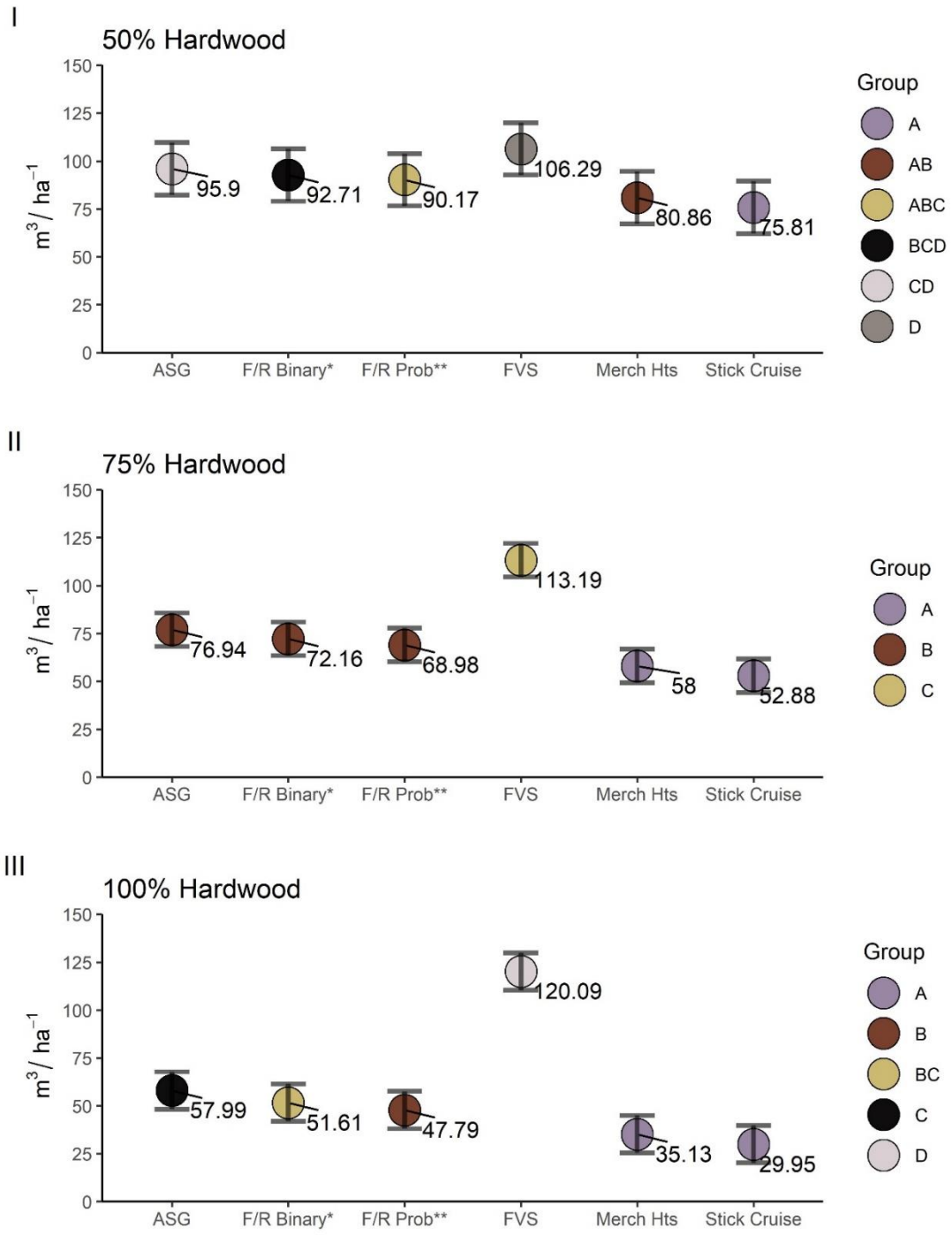


Figure 2.3 The predicted estimated marginal mean of plot level sawlog volume for four field merchandizing methods and FVS across three different forest types.

There was a statistically significant difference in observed plot-level sawlog volume among methods ($p<0.0001$) (Figure 2.4). FVS had the highest predicted sawlog volume of 112.96 ± 3.88 m^3/ha^{-1}. This was more than twice the value estimated using merchantable stopper heights and stick cruising, which showed no statistical difference from each other and had mean values of 38% and 34% of the FVS mean value, respectively. ASG had a predicted mean sawlog volume that was 56% of the mean FVS sawlog volume, while the probabilistic form and risk model had a mean predicted sawlog volume that was 48% of the mean FVS sawlog volume. The two-step binary form/risk model did not demonstrate significant difference from either ASG or the probabilistic model and had a plot-level value of 51.2% of the FVS sawlog volume, on average. This binary model used a cutoff value for predicting sawlogs that was optimized to balance errors of omission and commission equally for all species, and the results could be modified by using either a more or less conservative cutoff value for sawlog presence predictions.

Estimates of Merchantable Value

Tukey's honest significant difference test demonstrated meaningful statistical differences in plot-level financial value across methods ($p<0.0001$). FVS provided significantly higher estimates than all other methods examined with a mean value of \$5,616 ± \$186/ha^{-1} (Figure 2.5). ASG and both calculations of form/risk (i.e., the probabilistic model and the two-stage binary model) provided statistically similar estimates of 64%, 59%, and 61% of the FVS estimates, respectively. Merchantable stopper heights and stick cruising provided statistically similar values that were lower than ASG and both form/risk models. Merchantable stopper heights and stick cruising had mean plot-level values that were 48% and 46% of the FVS estimates, respectively.

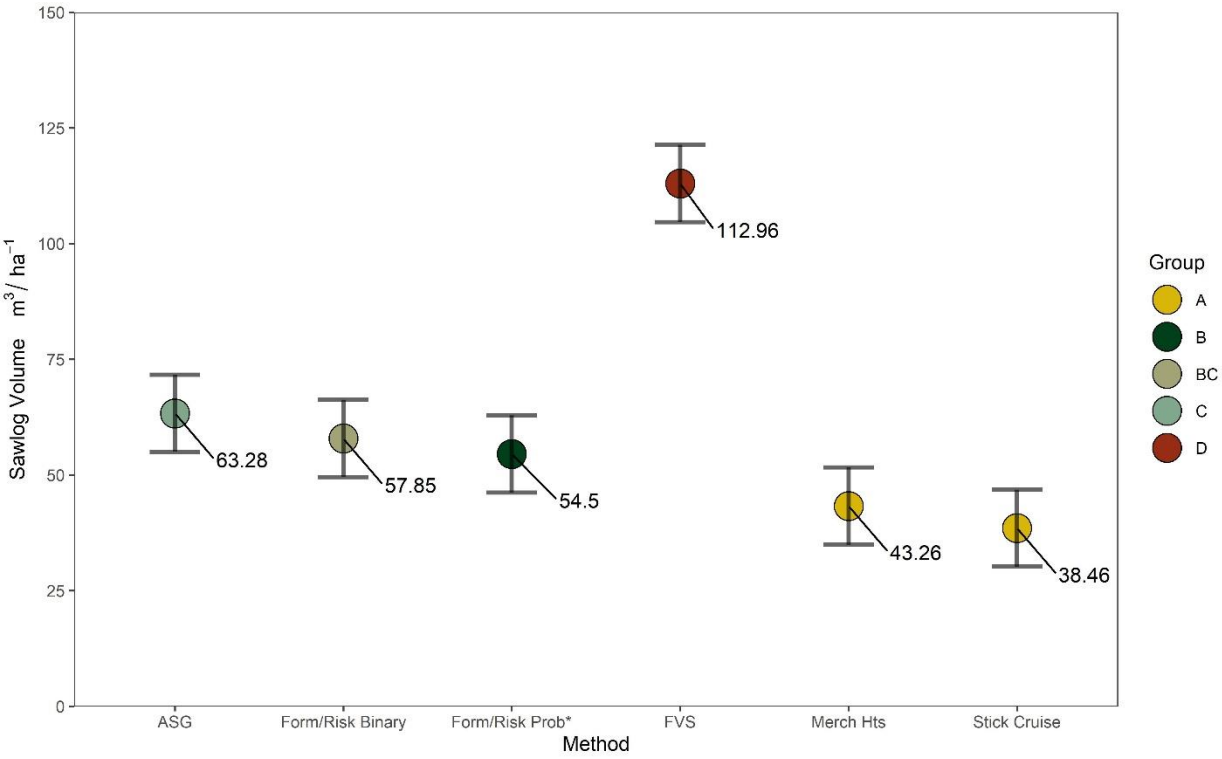


Figure 2.4 The estimated marginal mean value of meaured plot-level sawlog volumes for the four merchandizing methods including ASG, Merchantable Heights, Stick Cruising, Form/Risk, and the regional growth model FVS. The results of a two-stage binary model and a probabilistic model for form and risk predictions are displayed. Statistically significant differences in estimated means are differentiated by group.

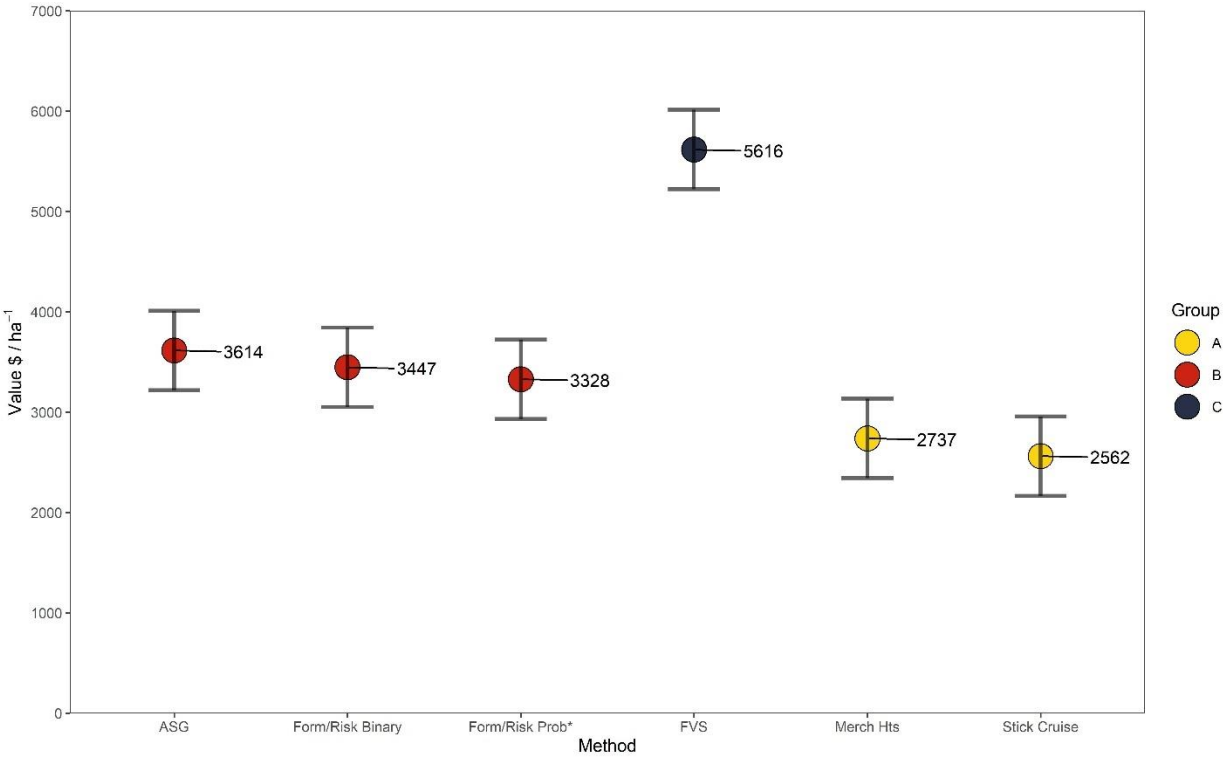


Figure 2.5 The estimated marginal mean value of stumpage values in $/ha^{-1} prior to consideration of the cost of harvesting or transportation for the four merchandizing methods ASG, Merchantable Heights, Stick Cruising, Form/Risk, and the regional growth model FVS. The results of a two-stage binary model and a probabilistic model for form/risk predictions are displayed. Statistically significant differences in estimated means are differentiated by group.

Variance In Merchantable Volume and Sawlog Volume Estimates

The mean observed coefficients of variation (CV) for value and sawlog volume when all measured plots were included for each method were 70.9% and 96.3%, respectively. All methods had a high degree of variability in the plot-level value measurements within each region (Table 2.5). The total merchantable volume had a CV of 47.5% when all measured plots were included. The CV for sawlog volume was between 1.82 times and 2.39 times more variable than merchantable volume for the field merchandizing methods tested. Sawlog volume was 1.14 times more variable for the generic FVS output than for merchantable volume. Regional values for sawlog volume and stand value were significantly higher in FVS outputs than any field merchandizing methods (Table 2.6).

Table 2.5 Coefficient of variation in measured merchantable volumes, sawlog volumes, and stumpage values for the five merchandizing methods including ASG, Merchantable Heights, Stick Cruising, Form/Risk, and FVS. The results of a two-stage binary model and a probabilistic model for form/risk predictions are displayed.

Method	**Northern Maine - Ashland**			**Western Maine - Rangeley**		
	Merchantable Volume CV	Sawlog Volume CV	Standing Value CV	Merchantable Volume CV	Sawlog Volume CV	Standing Value CV
FVS	48.4	67.7	68.2	46.6	47.7	50.1
Merch Hts	47.5	94.2	70.7	46.6	122.1	61.5
Stick Cruise	47.5	93.6	70.0	46.6	131.3	61.5
Form/Risk P	47.5	83.4	69.8	46.6	96.3	52.8
Form/Risk B	47.5	83.8	71.7	46.6	93.4	53.1
ASG	47.5	85.9	74.0	46.6	94.0	62.0

Table 2.6 Total measured values by region for merchantable volumes, sawlog volumes, and stumpage values for the five merchandizing methods including ASG, Merchantable Heights, Stick Cruising, Form/Risk, and FVS. The results of a two-stage binary model and a probabilistic model for form/risk predictions are displayed.

Method	Northern Maine - Ashland				Western Maine – Rangeley			
	Merchantable Volume (m^3)	Sawlog Volume (m^3)	Sawlog Volume %	Standing Value $	Merchantable Volume (m3)	Sawlog Volume (m^3)	Sawlog Volume %	Standing Value $
FVS	7,486	4,427	59.1	222,076	15,096	9,542	63.2	491,632
Merch Hts	8,073	1,959	24.3	115,194	16,174	3,534	21.8	235,911
Stick Cruise	8,073	1,799	22.2	110,391	16,174	3,045	18.8	218,581
Form/Risk P	8,073	2,432	30.1	140,792	16,174	4,529	28.0	284,667
Form/Risk B	8,073	2,570	31.8	145,380	16,174	4,824	29.8	295,067
ASG	8,073	2,761	34.2	149,493	16,174	5,318	32.9	311,944

Evaluating Efficiency of Inventory Methods

When the raw efficiency value was calculated for each method, we observed that merchantable stopper heights had the highest mean efficiency value and thus was the least efficient method (Figure 2.6). Form/risk showed no statistical difference from ASG or merchantable heights, while ASG showed no statistical difference from form/risk or stick cruising. Overall, stick cruising had the lowest mean efficiency score, suggesting it was the most efficient method. High standard error due to greater variability in the amount of time it took to perform each method across plots made it difficult to assess meaningful differences in the efficiency among methods.

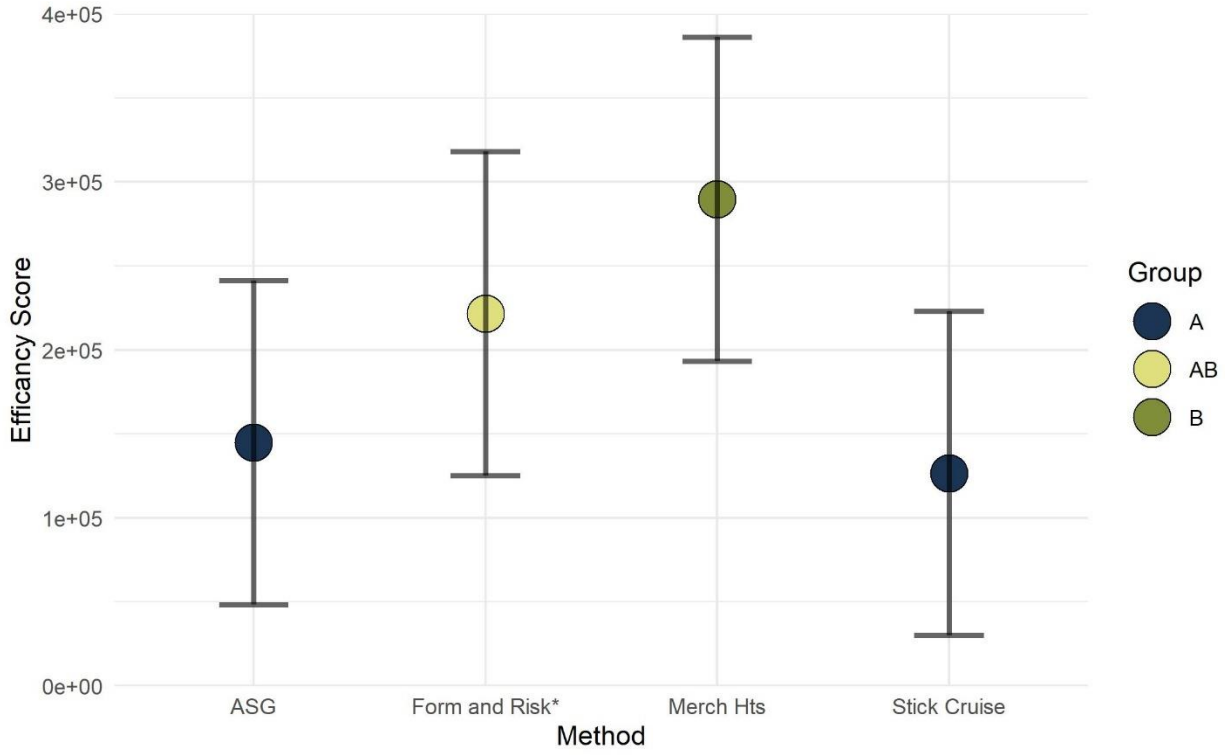


Figure 2.6 The estimated marginal mean efficiency for the four merchandizing methods including ASG, Merchantable Heights, Stick Cruising, Form/Risk, and FVS. The results of a two-stage binary model and a probabilistic model for form/risk predictions are displayed. Statistically significant differences in estimated means are differentiated by group.

Discussion

This analysis examined contrasting inventory methods for determining the merchantability of standing trees and their financial value for typical managed mixed-species stands in Maine. Statistically significant differences in estimates of both sawlog volume and stand value were observed, while the efficiency of methods was highly variable ($p < .05$). Although stick cruising and merchantable stopper heights provided estimates of sawlog volume that were not significantly different from each other, ASG, the probabilistic form/risk approach, and FVS all differed in their estimates of sawlog volume. The two-stage binary form/risk approach was not significantly different from ASG and the probabilistic form/risk approach. This was highly influenced by the cutpoint value used to discriminate between the occurrence of sawlog presence, and thus could be adapted to achieve different results depending on landowner-specific objectives.

All methods demonstrated sawlog volume as HW% increased, with the exception of FVS. FVS only responded to merchantable volume, and the slope may have been a result of established management goals in the measured stands. That is, if the stands were not managed for hardwoods but instead had a greater composition of large spruce-fir stems, the slope would reflect those conditions. While it remains true that percentage of hardwoods remains a covariate that interacts with method in this analysis, it is possible that other stand types commonly found in the region would not display such a strong negative correlation between percentage of hardwoods and percentage of merchantable volume as sawlogs. FVS did not meaningfully adjust outputs based on species composition, while the actual field measurements uniformly demonstrated a negative correlation between increased HW% and sawlog volume, if at modestly different rates (Figure 2.3).

ASG and both calculated form/risk approaches provided statistically similar plot-level value estimates and were all significantly lower than the estimates provided by FVS ($p < .05$). In terms of efficiency, all methods were relatively similar, suggesting tradeoffs between observed variability and the time required to make the necessary measurements.

FVS, a distance-independent tree-level growth and yield model, is frequently used to estimate when a stand can be reentered after a harvest using future estimates of merchantable volume (George et al., 2022). Our results showed that merchantable volume estimates using form and risk were 52% lower than FVS outputs. Using the more conservative stick cruising method, sawlog volume was 66% lower than FVS estimates and 44% lower than FVS estimates when using the least conservative ASG method. This can significantly alter the timing of treatment and harvest plans in hardwood dominated stands when the regeneration of sawlog-quality trees is the primary goal.

In general, this analysis highlights the sensitivity of estimates of both sawlog volume and potential financial value to the inventory method utilized. While there are many factors that influence the estimations of standing value, the methods used in this analysis were designed to provide consistency for the purpose of comparisons. Forest inventory data and the future projections of such data using growth and yield models are significant factors of forest property valuation (Holopainen et al., 2010). Given the large differences observed, it is important to note the broader implications and potential advantages, or disadvantages of the various methods examined.

Although financial value is often of the largest interest to landowners, it is driven primarily by available markets, and ostensible differences in value might mask other important differences across methods. In contrast, estimates of sawlog volume are more directly comparable and showed even greater distinctions across methods than did financial value. Estimates of sawtimber products are important because sawtimber is more valuable to a landowner than other forest products (*The Economic Importance of Maine's Forest-Based Economy 2013*, 2013). This finding has significant implications for harvest planning and future yield projections, since forest planners are highly dependent on estimates of sawlog yields for determining the operability of a particular stand or its economic value, variables that both evolve over time. Consequently, consistent and robust estimates of sawlog volume that reflect the actual variability found in the forest are required for forest managers to

make more informed decisions. Limiting unnecessary operational entries and targeting forest stands that can meet sawlog demands are requirements for profitable operations that can also lead to improved ecological outcomes.

Sawlog Volume and Stand Value Estimates

The literature exploring the variance in estimates of sawlog volume and stand values among field merchandizing methods is limited; to our knowledge, no other study has addressed the issue in the way this analysis has. This study found clear differences in value and, in particular, in sawlog volume as estimated by different merchandizing methods. This is likely due to the fact that pulp volumes were generally high in all of the stands, which is typical in the region where over 70% of harvested hardwood volume is pulp quality. This reduces the effect of sawlog volume in stand value estimates enough to limit the potential statistical differences between ASG and the two different calculations of the form/risk method.

Key drivers of sawlog volume estimates were basal area per hectare and species composition, which are logical factors given density effects on stem form and species differences in merchantability. The volume estimates for the stick cruising and stopper height methods may have been conservative because observers focused more heavily on identifying defects than may be typical in a faster-paced industrial cruise. This is important, as stick cruising and merchantable stopper heights are always going to be subjective, even when sawlog specifications are clearly defined. What one cruiser may identify as a disqualifying defect may be seen by another cruiser as just small enough to not limit a tree's sawlog potential. Stick cruising and merchantable stopper heights require field-based subjective judgments to be made for all portions of the stem. This allows more room for differences between cruisers. While this study had consistent observations made by the same observer in all samples, the results may have been different with either a more or less conservative observer. For example, Westfall et al. (2016) found mean systematic bias of upper-stem diameter ocular measurements between -0.36 and -0.68 cm for

common hardwood and softwood species, respectively. This negatively influences volume estimates. To avoid this potential bias, Westfall et al. concluded that upper-stem diameter measurements needed to be ± 0.2 cm, which required the use of specific measurement tools rather than ocular determination (Westfall et al. 2016).

With form and risk assessment, we know that the volume of sawlog material that is present is dependent upon two primary factors, species and diameter (Castle et al., 2017). All of the stands in this analysis had a history of management to favor species that produce valuable sawlog products. A more significant difference between form and risk and ASG may have been observed in an unmanaged forest populated with less sugar maple and yellow birch and more beech, red maple, and other hardwood species that display lower volumes of sawlog material using the form and risk methodology.

Differences in estimates were clear between methods, but they also were most evident in stands that had a higher percentage of hardwood. Both merchantable stopper heights and stick cruising methods had estimates that were strongly negatively influenced by a higher percentage of total volume made of hardwoods in a stand. While this was controlled for in the models used to calculate the estimated marginal means for the statistical comparisons, the differences between form/risk and merchantable stopper heights or stick cruising may be less dramatic in stands with a species composition that included a higher volume of softwoods.

Variability in Sawlog Volume, Value, and Merchantable Volume Estimates

Estimates of stand value and sawlog volume demonstrated significantly more variability than estimates of total merchantable volume. This emphasizes the general difficulty in assessing sawlog volume in the field. In general, forest inventories seek to minimize the amount of uncertainty in their estimates with a higher sampling intensity. Given that adding plots for the purpose of estimating sawlog volume alone may not always be possible, the generally high variability observed among plots in this analysis highlights the importance of making field-based estimates of sawlog volume at all sample plot

locations. In addition, given the large differences in both sawlog volume and stand value, it is vital to select the right method for estimation and potentially measure at a higher sampling intensity than was done in this analysis, which needs to be taken into account when budgeting for field inventories, particularly those conducted pre-harvest when robust estimates of financial value are needed. Knowing the variability in sawlog estimates allows planners to accurately report potential values to other stakeholders.

It is up to individual organizations and land managers to assess the value of decreased error in sawlog volume estimates through more intensive sampling against the potential losses that might be incurred by poor or less informed decision-making. Knowing the variance and error rates allows for a more accurate cost-plus-loss analysis where the true cost of collecting high quality sawlog volume data can be measured against the losses from less informed decision making. This is an important analysis that cannot be conducted without reasonable estimates of error (Holopainen et al., 2010). Since variability is so much higher in estimates of sawlog volume than other volume estimates, planning inventories based on reducing error in the measurement of merchantable volume alone will not capture the variability of sawlog volume in multi-cohort managed stands with complex species composition like those included in this study.

Evaluation of Efficiency Across Field Merchandizing Methodologies

This study demonstrated that there were important and statistically significant differences in the efficiency of the methodologies included in this analysis. While these differences were present, the standard errors in efficiency were relatively high across methods, which makes generalized conclusions hard to make. This is further complicated by potential issues with the actual methodology for assessing efficiency. In a general field inventory, each methodology would be fully integrated into the entire inventory process, and not separated from the standard diameter or height measurements in the way they were in this study. For example, form/risk analysis could be conducted while DBH measurements

are being made, improving the method's efficiency compared to the results in this study. Additionally, both ASG and stick cruising are reliant on visual estimates, but the estimates were made on trees that had already been assessed during the measuring of DBH and height. This potentially artificially increased the efficiency of these two methods.

Advantages and Limitations of Field Merchandizing Methods

There are several key limitations and advantages to each merchandizing method that should be highlighted further. A primary limitation to the ASG, merchantable stopper height, and stick cruising methods is that results will be highly variable among cruisers, as sawlog presence and the estimated sawlog length are subjective assessments. The ability to recognize the importance of defects, visually estimate sweep, and make assumptions about internal defect and decay is based primarily on training, experience, and best judgment. In contrast, the form/risk method requires relatively simple observations based on an easy-to-approach field guide that does not require a cruiser to have extensive knowledge of tree scaling and log grading (Appendix D). This may lead to more consistent and accurate observations regardless of the cruiser.

Stick cruising and merchantable stopper heights provide the benefit of specifically identifying where on the stem that sawlog material is present, but the value of this information may decrease when fully integrated into a landscape-level inventory framework where the representative nature of these measurements is unknown. Although merchantable stopper height allows for a more precise and less demanding measurement than stick cruising, both methods were shown to have statistically similar estimates of plot-level value and sawlog volume. Form/risk and ASG methods do not provide information about the specific merchantable portions of a stem, while FVS provides no real additional information beyond what can be derived from merchantable volume estimates.

A primary challenge with FVS is understanding its defaults and ensuring a proper volume equation is being used for the specific location, something most general users of the regional model may not

realize. While FVS is widely used to compute current conditions and conduct future projections, estimates of stand-level variability are often not a default output and the model must be properly initialized to obtain this information.

A primary advantage of form and risk analysis is that it provides more than a simple static measure of condition and is linked to an extensive database of observations across a variety of conditions. Form/risk estimates adapt to changing diameters and heights for measured trees as well as changing stand conditions that alter a tree's relative competition within the stand through time. Merchantable height, stick cruising, and ASG provide static measurements. The ability of form and risk analysis to adapt to future conditions may make the method a more robust approach for estimating sawlog volume and economic value across a broad range of conditions if integrated into growth and yield projections. This is especially true given the clear overestimations provided by the current default FVS outputs.

Recommended Field Merchandizing Approach

This analysis has illuminated the significant differences in estimates of sawlog volume and value among field merchandizing methodologies and shown the higher variability in sawlog volume estimates when compared to merchantable volume. This analysis has also highlighted specific differences and limitations in the applications of the different methodologies. When considering the various tradeoffs among methods and the findings of this analysis, the form and risk approach was determined to be the best method for estimating sawlog volume and stand values due to reasonableness of estimates, reproducibility, relative efficiency, and the ability to modify predictions over time in growth and yield models as demonstrated by Castle et al. (2018). In this prior analysis, the use of stem form and risk observations reduced projections of stand value by over 17% on average (range of 13% to 28%) when compared with projections that did not include that tree-level information (Castle et al. 2018).

In general, all the examined field merchandizing methods produced different plot- and stand-level estimates of standing sawlog volume but provided estimates that were lower than those provided

by FVS. Although most of these estimates were statistically different from each other, they were generally reasonable across the broad range of conditions encountered. A true test of their accuracy was not possible in this study since a complete harvest with product tracking would have been required. Although the form and risk approach was not the most efficient method in this study, the form and risk classification system used in this analysis included four risk classes and eight form classes, which could be further simplified into two form classes and three risk classes as seen in chapter 1. In addition, the research design prevented assigning form and risk to the individual trees while taking height and diameter measurements, which limited efficiency. Both form and risk measurements are based on observable features that can be simplified into an easy-to-follow visual field guide that can remove most of the subjectivity from the assessment (Table 2.2, Figure 2.2). This means that observations are more likely to be consistent and reliable over time and space regardless of the skill level or experience of inventory crews. Given the relatively limited resources often dedicated to field inventories, the efficiency of measurement and specific field protocols is critical.

In contrast to stem form and risk assessment, stick cruising and merchantable stopper heights require well trained and experienced field crews to make subjective decisions about the likelihood that each tree be of sawlog quality. These methods also require visual estimation of the effect of various defects along parts of the stem above the first couple of meters, an effort that can be quite difficult. While ASG observations may be more consistent because they simply record the binary occurrence of sawlogs without requiring estimates of sawlog length, they too require a subjective decision concerning sawlog presence. Different cruisers are likely to have contrasting opinions and estimates on the same trees, which creates inconsistency in estimates. This is particularly true when inventories are taken over longer time scales with multiple field crews. This increased variability and subjectivity make it more difficult to estimate the amount of uncertainty that is present in the inventory data.

Given these advantages, the stem form and risk assessment method is likely the most effective inventory method for obtaining estimates of sawlog volume and stand values in the field. This is especially true for the purposes of growth and yield modeling as form and risk analysis is the only method that meaningfully adapts estimates to changing tree- and stand-level conditions over time.

Summary

Irrespective of the field methodology used, estimates of sawlog volume and merchantable value show a high degree of variability with a mean CV of 96% and 64% across plots, respectively. All the field methods examined provided statistically significant ($p < .001$) estimates of sawlog volume and merchantable value that were lower in value and higher in variability when compared to those from FVS. This study highlights that a higher degree of sampling intensity is needed to reduce error in measurements of sawlog volume and merchantable value in complex multi-cohort managed hardwood stands with mixed species composition like those included in this analysis.

Understanding that the field inventory method used to estimate sawlog volume and merchantable value affects estimates in significant ways, this study was unable to assess the actual accuracy of each methodology. Further research is needed to compare predicted and realized sawlog yields using each methodology to better determine which is the best one to use across a range of conditions. Additionally, while attempts to reduce bias in measurements of efficiency were made, the measurements were still heavily biased by the observations made measuring DBH and height and the effect that these potentially had on stick cruising and ASG efficiency. Additionally, since the merchandizing methodologies would likely be integrated into an inventory in a more coherent way than was done in this study, our measures of efficiency did not measure the efficiency of these methods in a practical application.

Overall, this study has shown the importance of considering the benefits and limitations of different field inventory methods for estimating sawlog volume and merchantable value. This is

particularly important in operations that optimize their harvesting plans based on sawlog volume and estimated net present value. In short, estimating sawlog volume and merchantable value in complex multi-cohort managed forests is a difficult task, and the resources dedicated must be considered when there is concern about these values. This study demonstrates that further research needs to be done on the topic to understand and select the tools available in accomplishing this task most effectively.

EPILOGUE

Stem form and risk in hardwood species and certain conifers such as Eastern hemlock and Eastern white pine is a complex phenomenon that has proved to be both easy to measure and difficult to predict. These field measurements provide a powerful tool for predicting in-field determinations of sawlog presence and volume. The distribution of stem forms varies dramatically between species. This is not surprising given species' different growth and regeneration strategies and the complex dynamics of a naturally regenerating hardwood dominant forest.

The diversity of stem forms and their risk as defined in this study have been shown to be strong predictors of in-field determinations of sawlog presence and the percentage of merchantable volume available as sawlog material. This fact, in addition to the role stem form and risk have been shown to play in diameter growth and mortality in Castle et al. (2018), demonstrates how important these observations may be for not only capturing present conditions but also modeling future growth and yield.

Static measurements of sawlog volume certainly have value in assessing current stand value and operability, but making estimates of the value of standing timber over time requires a dynamic estimation of product potential in future growth. Using form and risk as a framework for estimating sawlog volume in growth and yield modeling allows for estimates to adapt to predicted changes in stand conditions. Additionally, form and risk assessment has shown to provide reasonable estimates of sawlog volume and stand value when compared to other commonly used field merchandizing methods. The predicted sawlog volume was especially reasonable when compared to the standard outputs from the regional growth model FVS.

Limitations

There were limitations to the work done in this study and the potential reliability of conclusions. First, in chapter one, when looking at form and risk analysis as a field merchandizing method we were

not able to establish that tree-level form and risk classes are representative of site-, stand- or stratum-level conditions or if they are tree-specific and unable to capture stand-level dynamics. This is true for all the inventory methods tested in chapter two but is a limitation that needs to be considered when applying the methodology to estimate standing sawlog value across large ownerships.

All sample locations included in chapters one and two of this study are in managed forests with an extensive history of harvesting for timber products. It is unclear if the results of this study would be similar in unmanaged stands. Chapter one of this study was also unable to make more detailed species-specific observations concerning the prediction of stem form, risk class, sawlog likelihood, and sawlog volume due to a lack of data. This is particularly true for the sawlog likelihood and sawlog volume models. The models are likely less robust in their predictions for cedar, white pine, and minor hardwood species in the region due to the sample size.

Additionally, the models in chapter one were derived from field observations and not trees that were felled and sawn for lumber. While the observers were consistent in their methodology, actual product recovery was not used. Trees were not harvested and then tracked to the mill; thus, the model estimations may be optimistic. As previously discussed, it is difficult to know the accuracy of product estimates in standing timber.

In chapter two there are two significant limitations. First is the inability to measure the efficiency of each method. The research design was unable to capture relative efficiency due to the fact that the same individuals were merchandizing each tree several times. While efforts were made to randomize the order of methods and standardize the approach, standard errors were too high to draw meaningful conclusions and the author believes that the measures of efficiency are not reflective of actual efficiency when each of these methods are implemented in practical application. The second limitation is that many of the merchandizing methods require subjective measurements. It is possible

that different cruisers would arrive at different estimates when using merchantable stopper heights, stick cruising and ASG methodologies. This highlights the problems inherent to these specific methodologies, and while specific protocols were established to define sawlog potential, if different cruisers were to follow the same study design, they may have come to different conclusions at the tree level about sawlog presence and volume. Utilizing a diversity of experienced timber cruisers may help either highlight or resolve this issue.

Management Implications

The use of form and risk analysis in inventories has several management implications in hardwood dominant forests. Given that these forests are managed for sawlog products, having a tool to predict current and future sawlog volume can influence the timing of stand improvement actions and harvest schedules. Identifying a stand's prime rotation age using sawlog volume instead of total merchantable volume can change determinations of when a stand is operable and how frequent operational entries should be using selection harvest silvicultural techniques that are common in the hardwood dominant forest type studied. This can improve both economic and ecological outcomes.

The form and risk equations also highlight the significant role species composition plays in determining stand operability. All species had a decreasing percentage of sawlog volume as a percentage of merchantable volume at a species-specific diameter. This demonstrates the fact that quadratic mean diameter and total merchantable volume estimates have different implications for the management of different species and stand types. This is important when long term planning uses basal area or merchantable volume as a trigger for management decisions when projected sawlog volume may be a more effective trigger that produces different results.

Silvicultural decisions for stands dominated by northern white-cedar and stands dominated by sugar maple will be different due to different site conditions, species regeneration strategies, and other

ecological factors. What form and risk analysis can do is help prevent operational entry into such stands before they have reached their peak sawlog potential due to form and risk distribution and other factors such as diameter distribution and stand density. It can also help time harvests to make sure that value is not lost on the stump due to declining sawlog potential. Peak sawlog potential is going to be reached in different conditions for each forest type and these conditions are something that can be effectively modeled. This may have the effect of increasing economic outcomes by reducing the number of entries that do not improve stand health and that result in the excessive extraction of low value pulp products.

Another potential implication of the measurement of form and risk is in the management of spruce-fir stands, given the relative accuracy of the models developed for predicting form and risk in red spruce and balsam fir. This provides a way to more accurately predict cull volumes that otherwise would be difficult without predicting form and risk class. New technologies like remote sensing can give important information about stem density and volume, but not stem quality. Form and risk allow for the prediction of the likelihood of disqualifying defects associated with non-ideal stem form and can be used to make more robust estimates of the ability of spruce-fir species to produce sawlogs. For large landowners, a small increase in the accuracy of sawlog predictions for spruce and fir species can have significant economic consequences. This also has the potential to inform effective resource allocation by identifying ideal site and stand conditions for high quality sawlogs.

Chapter two demonstrates the importance of the inventory methodology chosen for estimating sawlog volume. Additionally, the Forest Vegetation Simulator's default outputs were shown to provide significantly higher estimates than any field merchandizing method. It is reasonable to assume that implementing a field merchandizing method will provide significantly more accurate estimates of standing sawlog volume. There are often limited resources available for forest inventory, and utilizing efficient, effective, and reproducible methods for forest measurement are important. This is an important consideration for inventory planners. The benefits of utilizing form and risk as a method were

expanded upon in chapter two. A reproducible and reasonably efficient method that gives robust estimates of sawlog volume and that adapts to changing conditions over time is a powerful tool that will give forest planners a significantly better understanding of their land base. This increased understanding has the potential to lead to better economic and ecological outcomes through the more effective timing of stand improvements and harvests.

Managing hardwood forests is a difficult task. There is a constant balance between economic needs, ecological concerns, forest health, encouraging regeneration of desired species, and high variability among and within stands. This work is hopefully the beginning of thinking about form and risk to prioritize stands for harvest or stand improvement, increase economic outcomes, and if further data is collected and research pursued – improve future projections of growth, mortality, and estimates of product ratios. While this work only attempts to examine the latter, form and risk may be a powerful framework for improving the quality of and insight gained from field inventories in hardwood dominate forests.

Future Research

In this research, stem form and risk class has proved itself to be a powerful framework for improving predictions of standing uncut sawlog volume in hardwoods, spruce-fir, and several other conifer species. Prior research has demonstrated that the same stem form and risk framework can be a powerful tool for improving diameter growth and mortality equations in hardwood and mixed-hardwood forests commonly found in Maine and the entire northeastern region of the United States. This leaves several potential opportunities for future research.

- The continued collection of field measurements can refine sawlog volume predictions and sawlog likelihood predictions. Additional data may allow the creation of more accurate

species-specific models that give land managers improved tools for making informed management decisions.

- Additional data may allow for a more sophisticated breakdown of the role of each individual form class, or different groupings of form classes, in potential product recovery. This study identified either ideal or non-ideal form as significant, but it is possible that different stem forms are significant either on a species-specific level or, with more data, across all species.
- Ongoing research with continuous forest inventory plots to better track the relationship between stem form and risk in diameter growth, mortality, and sawlog presence over time could allow for fitting of better models that more accurately predict mortality, growth, and sawlog likelihood in complex mixed species hardwood and mixed hardwood stands.
- Additional analysis using larger data sets can identify if form and risk class can be more accurately predicted using species-specific models. It is possible stem form and risk may prove to be too heavily influenced by management history and other sources of site-specific variability that are hard to capture. It is also possible that when species are analyzed separately, site-level variables, height, diameter, stand density and relative social position may be effective predictors of stem form that can capture each species' preferred site conditions and regeneration strategies.

Additional research is necessary to assess the accuracy of each inventory method that was tested in chapter two. There are many opportunities for future research in this topic.

- While we were able to recognize the differences in sawlog volume and stand-level value predictions using each method, the absolute accuracy was not assessed. Studying the accuracy of each method based on actual product recovery after harvest would allow land

managers to determine which inventory methodology is most effective for their measurement protocols.

- A follow-up study using different cruisers and the same methodologies may highlight how significant the role of subjective judgements was in field merchandizing.
- Comparing the field estimates using each method for several cruising teams across the same research plots may provide insight into the reproducibility of each method when applied in the field. This can include subsets of experienced cruisers and entry level cruisers and would address the issue of consistency and bias in field merchandizing measurements for each methodology.
- Finally, a study that assesses differences in predicted standing value of uncut timber using more sophisticated product and species-specific merchandizing software would provide a clearer picture of how much each method affects estimates of stand values.

Summary

This study provides a baseline for assessing both the importance of form and risk for merchandizing during field inventories and the importance of understanding the tradeoffs land managers make when choosing a field merchandizing methodology. While there were limitations, meaningful management implications and avenues for further research were identified. The proposed form and risk methodology provides a step forward in the improvement of field merchandizing inventories both for short-term harvest assessments and long-term growth and yield modeling. This is a subject where research has been scarce to non-existent, and there remains room both for the development of better and more accurate uses of the form and risk observations and for the improvement of field merchandizing techniques. This includes using form/risk for sawlog volume estimates as well as diameter growth and mortality over time. This study provides a baseline for future

work that improves the ability of land managers to plan inventories, make management decisions in hardwood forests, and estimate current and future standing sawlog volume.

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APPENDIX

A – Supplementary Tables

Table A.1 Parameter estimates with associated standard errors (SE) and fit statistics for equations 1 and 2.

	Eq. 1 (Binomial) – Hardwood Ideal Form				Eq. 2 (Binomial) Hardwood Risk Class 1			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 0$	10.2075	1.3312	<.001	$\beta 20$	5.1004	2.3168	.027
HT	$\beta 1$	0.2268	0.0242	<.001				
I(HT/DBH) Tallest					$\beta 21$	3.2365	0.3339	<.001
BAL								
DD5	$\beta 2$	-0.0077	0.0007	<.001				
FFP					$\beta 22$	-0.0984	0.0190	<.001
OH	$\beta 3$	-0.5678	0.5368	.290	$\beta 23$	4.7368	0.5322	<.001
PB	$\beta 3$	-0.2259	0.5066	.656	$\beta 23$	5.3648	0.5048	<.001
QA	$\beta 3$	2.9417	0.5744	<.001	$\beta 23$	6.1257	0.5599	<.001
RM	$\beta 3$	-1.1916	0.4580	.009	$\beta 23$	4.0752	0.4506	<.001
RO	$\beta 3$	-1.2560	0.4987	.012	$\beta 23$	4.9593	0.4853	<.001
SM	$\beta 3$	0.2249	0.5000	.653	$\beta 23$	5.2356	0.4756	<.001
WA	$\beta 3$	-2.0400	0.7629	.008	$\beta 23$	5.9764	0.7474	<.001
YB	$\beta 3$	-1.6315	0.4751	<.001	$\beta 23$	4.6847	0.4651	<.001
Fit Statistics								
AUC (FE, RE)		(.67, .78)	(.01, .01)			(.66, .76)	(.01, .01)	

Table A.2 Parameter estimates with associated standard errors (SE) and fit statistics for equations 3 and 4.

	Eq. 3 (Binomial) Hardwood Risk Class 2				Eq. 4 (Binomial) Hardwood Risk Class 3			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 30$	-16.4301	1.7821	<.001	$\beta 40$	-1.5934	0.5806	.006
log(HT/DBH)					$\beta 41$	-2.2437	0.2719	<.001
I(Tallest/HT)					$\beta 42$	1.6271	0.3206	<.001
BAL	$\beta 31$	-0.0286	0.0641	<.001	$\beta 43$	0.0146	0.0087	<.001
FFP	$\beta 32$	0.1042	1.4976	<.001				
OH	$\beta 33$	0.9913	0.5085	.051	$\beta 44$	-5.3222	0.4091	<.001
PB	$\beta 33$	0.7470	0.4777	.118	$\beta 44$	-5.5906	0.3937	<.001
QA	$\beta 33$	-0.4485	0.5221	.390	$\beta 44$	-5.2242	0.4495	<.001
RM	$\beta 33$	1.2499	0.4273	.004	$\beta 44$	-4.8419	0.3489	<.001
RO	$\beta 33$	2.4248	0.4637	<.001	$\beta 44$	-6.6592	0.3781	<.001
SM	$\beta 33$	1.1371	0.4525	.012	$\beta 44$	-5.7748	0.3781	<.001
WA	$\beta 33$	-0.5918	0.7150	.408	$\beta 44$	-4.6448	0.5864	<.001
YB	$\beta 33$	1.3419	0.4426	.002	$\beta 44$	-5.7313	0.3615	<.001
Fit Statistics								
AUC (FE, RE)		(.67, .73)	(.01, .01)			(.69, .82)	(.02, .01)	

Table A.3 Parameter estimates with associated standard errors (SE) and fit statistics for equations 5 and 6.

	Eq. 5 (Binomial) Spruce-Fir Ideal Form				Eq. 6 (Binomial) Spruce-Fir Risk Class 1			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 50$	15.1476	0.9605	<.001	$\beta 60$	54.4001	3.7656	<.001
DBH	$\beta 51$	-0.0229	0.0078	.003				
log(DBH)					$\beta 61$	2.3878	0.2079	<.001
MWMT	$\beta 52$	-0.3877	0.0411	<.001	$\beta 62$	-2.7382	0.1672	<.001
MAP	$\beta 53$	-0.0010	0.0002	<.001	$\beta 63$	-0.0073	0.0007	<.001
RS	$\beta 54$	-0.4415	0.1929	.022	$\beta 64$	0.9489	0.1597	<.001
RS x DBH	$\beta 55$	0.0247	0.0088	.005				
Fit Statistics								
AUC (FE, RE)		(.72, .78)	(.03, .03)			(.89, .94)	(.01, .01)	

Table A.4 Parameter estimates with associated standard errors (SE) and fit statistics for equations 7 and 8.

	Eq. 7 (Binomial) Spruce-Fir Risk Class 2				Eq. 8 (Binomial) Spruce-Fir Risk Class 3			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 70$	-17.3724	1.5096	<.001	$\beta 80$	-12.1762	1.3238	<.001
log(DBH)	$\beta 71$	-2.4075	0.1767	<.001				
MAP	$\beta 72$	0.0037	0.0006	<.001				
DD5	$\beta 73$	0.0092	0.0006	<.001	$\beta 81$	0.0045	0.0009	<.001
CMD					$\beta 82$	-0.0091	0.0036	.011
RS					$\beta 83$	-0.6504	0.1102	<.001
Fit Statistics								
AUC (FE, RE)		(.88, .91)	(.01, .01)			(.72, .89)	(.02, .01)	

Table A.5 Parameter estimates with associated standard errors (SE) and fit statistics for equations 9 and 10.

	Eq. 9 (Binomial) Other Softwoods Ideal				Eq. 10 (Binomial) Other Softwoods Risk 1			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 90$	54.9161	3.7182	<.001	$\beta 100$	62.4387	5.1596	<.001
log(DBH)					$\beta 101$	4.2624	0.2185	<.001
log(HT/DBH)	$\beta 91$	-0.5261	0.2206	.017				
log(Tallest)					$\beta 102$	-5.2769	1.1661	<.001
MWMT					$\beta 103$	-3.0239	0.2398	<.001
Eref	$\beta 92$	-0.0724	0.0053	<.001				
OC	$\beta 93$	-3.0304	0.3605	<.001	$\beta 104$	-2.1065	0.4156	<.001
OS	$\beta 93$	-1.2725	0.5789	.028	$\beta 104$	-0.6839	0.7201	.293
WP	$\beta 93$	0.0782	0.2103	.710	$\beta 104$	-1.4565	0.2513	<.001
Fit Statistics								
AUC (FE, RE)		(.65, .84)	(.02, .02)			(.76, .85)	(.01, .01)	

Table A.6 Parameter estimates with associated standard errors (SE) and fit statistics for equations 11 and 12.

	Eq. 11 (Binomial) Other Softwoods Risk 2				Eq. 12 (Binomial) Other Softwoods Risk 3			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 110$	11.1089	1.6408	<.001	$\beta 120$	-7.8893	1.2974	<.001
log(DBH)	$\beta 111$	-5.0918	0.3313	<.001	$\beta 121$	-0.9551	0.2441	<.001
BAL					$\beta 122$	0.0452	0.0095	<.001
BMGI	$\beta 112$	0.0008	0.0003	.013	$\beta 123$	0.0008	0.0003	.001
OC	$\beta 113$	-11.3161	2.4917	<.001	$\beta 124$	2.5560	0.2950	<.001
OS	$\beta 113$	-20.8059	5.0669	<.001	$\beta 124$	1.4271	0.5015	.005
WP	$\beta 113$	-14.8261	1.4336	<.001	$\beta 124$	2.7920	0.1918	<.001
OC x log(DBH)	$\beta 114$	2.8012	0.7508	<.001				
OS x log(DBH)	$\beta 114$	5.8662	1.5830	<.001				
WP x log(DBH)	$\beta 114$	4.1768	0.4317	<.001				
Fit Statistics								
AUC (FE, RE)		(.69, .78)	(.02, .01)			(.74, .86)	(.02, .02)	

Table A.7 Parameter estimates with associated standard errors (SE) and fit statistics for equations 17 and 18.

	Eq. 17 (beta) - Spruce-Fir Svol/Mvol				Eq. 18 (beta) – Other Softwoods Svol/Mvol			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 170$	-27.5014	0.6721	<.001	$\beta 180$	-25.1200	5.0210	<.001
log(DBH)	$\beta 171$	12.2515	0.3337	<.001	$\beta 181$	9.7521	1.8250	<.001
Log(HT/DBH)								
Form1	$\beta 172$	0.4646	0.1512	.002				
Risk3					$\beta 182$	-0.3022	0.2262	.148
CSI					$\beta 183$	-0.0943	0.0459	.037
BF x DBH	$\beta 173$	-0.3671	0.0174	<.001				
RS x DBH	$\beta 173$	-0.3651	0.0171	<.001				
OS x DBH								
EH x DBH					$\beta 184$	-0.1636	0.0369	<.001
OC x DBH					$\beta 184$	-0.1752	0.0380	<.001
WP x DBH					$\beta 184$	-0.1744	0.0351	<.001
Fit Statistics								
R2		.79				.39		
RMSE		0.0617				0.1796		
MAE		0.0252				0.1140		
Bias		-0.0028				-0.0442		

Table A.8 Parameter estimates with associated standard errors (SE) and fit statistics for equations 14 and 15.

	Eq. 14 (Binomial) Spruce-Fir Has Sawlog				Eq. 15 (Binomial) Other Softwoods Has Sawlog			
Ind. Variable	Par.	Estimate	SE	p-value	Par.	Estimate	SE	p-value
Intercept	$\beta 140$	-55.4585	4.3643	<.001	$\beta 150$	-79.3273	8.0457	<.001
log(DBH)	$\beta 141$	23.3889	2.1257	<.001	$\beta 151$	24.8516	3.6384	<.001
Risk2	$\beta 142$	-3.6702	0.6626	<.001	$\beta 152$	-1.4568	0.8860	.103
Risk3	$\beta 142$	-9.5663	0.6336	<.001	$\beta 152$	-4.6883	0.9626	<.001
log(HT)	$\beta 143$	3.2084	0.9879	.001				
CSI					$\beta 153$	0.3641	0.1442	.013
BF x DBH	$\beta 144$	-0.8407	0.0877	<.001				
RS x DBH	$\beta 144$	-0.8472	0.0835	<.001				
OS x DBH					$\beta 154$	-0.8246	0.1642	<.001
EH x DBH					$\beta 154$	-0.6503	0.1021	<.001
OC x DBH					$\beta 154$	-0.6121	0.1044	<.001
WP x DBH					$\beta 154$	-0.5534	0.0814	<.001
OS x log(HT)					$\beta 155$	6.2570	2.2138	.006
EH x log(HT)					$\beta 155$	4.1018	2.0928	.053
OC x log(HT)					$\beta 155$	2.9770	2.0957	.158
WP x log(HT)					$\beta 155$	3.4551	2.0391	.093
Fit Statistics								
AUC (FE, RE)		(.90, .94)	(.03, .02)			(.88, .97)	(.04, .02)	

Table A.9 Summary of NHRI form class and risk class distributions for modeled species.

Species	Risk Class				Form Class								Total Trees
HARDWOOD SPECIES	**R1**	**R2**	**R3**	**R4**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**	
American Beech	42	52	124	51	133	59	10	34	13	6	7	7	**269**
Paper Birch	423	168	32	32	293	221	14	6	19	23	10	69	**655**
Quaking Aspen	352	91	26	31	316	134	0	1	4	4	0	41	**500**
Red maple	1311	1132	370	212	1075	534	26	13	178	51	87	1061	**3025**
Red Oak	629	975	87	37	667	589	6	0	94	119	60	193	**1728**
Sugar Maple	577	269	144	65	750	166	36	5	15	26	31	26	**1055**
White Ash	77	35	14	3	53	58	1	0	3	2	5	7	**129**
Yellow Birch	625	352	149	66	584	248	76	49	60	25	44	106	**1192**
Other Hardwoods	206	159	49	24	188	162	1	2	25	34	7	19	**438**
TOTAL OBSERVATIONS	**4242**	**3233**	**995**	**521**	**4059**	**2171**	**170**	**110**	**411**	**290**	**251**	**1529**	**8991**
SOFTWOOD SPECIES	**R1**	**R2**	**R3**	**R4**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**	
Balsam Fir	1742	808	184	91	2719	55	0	0	8	12	12	19	**2825**
Red and White Spruce	1728	421	77	27	2193	32	0	0	4	2	15	7	**2253**
Eastern Hemlock	881	793	83	30	1396	233	1	5	42	32	28	50	**1787**
Northern white-cedar	513	101	79	27	520	119	1	1	8	43	13	15	**720**
Eastern White Pine	853	619	353	131	1582	256	10	7	31	4	40	26	**1956**
Other Softwoods	103	27	9	4	110	11	0	1	6	2	11	2	**143**
TOTAL OBSERVATIONS	**5820**	**2769**	**785**	**310**	**8520**	**706**	**12**	**14**	**99**	**95**	**119**	**119**	**9684**

Table A.10 Summary of maximum sawlog volumes, maximum probabilities and 50% probability of sawlog diameters for all modeled species.

Species	**Max S_{vol}/M_{vol}**	**Max S_{vol}/M_{vol} DBH (cm)**	**50% S_{vol} / M_{vol} (Min DBH) (cm) (Max DBH) (cm)**	**Max pr(Sawlog) DBH (cm)**	**50% pr(Sawlog) (Min DBH) (cm) (Max DBH) (cm)**
Paper Birch	71.1%	46.1	(31.2) (65.0)	46.1	(31. 2) (65.0)
Red Maple	72.2%	46.3	(33.2) (65.9)	46.2	(31.0) (65.9)
Red Oak	72.7%	46.3	(30.9) (66.3)	46.3	(30.8) (66.3)
Sugar Maple	73.6%	46.5	(30.7) (67.1)	46.5	(30.6) (67.1)
White Ash	82.5%	48.3	(28.7) (75.3)	48.3	(28.7) (75.3)
Yellow Birch	75.0%	46.8	(30.3) (68.2)	46.3	(30.3) (68.2)
White Pine	96.3%	59.6	(23.5) (122.4)	40.4	(23.2) (64.5)
White Cedar	93.0%	55.7	(24.3) (106.6)	29.6	(13.7) (54.6)
Eastern Hemlock	93.3%	55.9	(24.2) (107.6)	37.9	(18.9) (66.8)
Red Spruce	97.9%	33.9	(13.7) (68.2)	27.8	(10.5) (58.0)
Balsam Fir	98.1%	34.2	(13.6) (69.5)	27.6	(10.6) (57.2)

Table A.11 Merchantable volume and stumpage value with coefficient of variation before consideration of harvesting and transportation costs for all measured stands.

Estimated Stand Values and Variability														
	Merchantable Volume		Form/Risk**		Form/Risk*		Stick Cruising		Acceptable Sawlog Grade		Merchantable Stopper Heights		Forest Vegetation Simulator	
Stands	Total	CV	Value	CV	Value	CV	Value	CV	Value	CV	Value	CV	Value	CV
1A	1541.51	44.0%	$24,696	77.0%	$26,876	74.0%	$20,357	62.0%	$32,016	54.0%	$21,817	66.0%	$39,398	65.0%
2A	1800.72	29.0%	$35,192	51.0%	$36,961	44.0%	$26,733	60.0%	$35,835	33.0%	$27,105	63.0%	$54,500	39.0%
3A	1315.50	46.0%	$24,879	83.0%	$22,559	88.0%	$18,318	83.0%	$25,989	68.0%	$18,926	84.0%	$34,099	72.0%
4A	1948.42	54.0%	$34,362	66.0%	$37,259	71.0%	$23,364	71.0%	$29,171	78.0%	$25,280	70.0%	$56,380	89.0%
5A	1343.85	53.0%	$21,539	87.0%	$21,529	90.0%	$21,008	69.0%	$27,845	66.0%	$21,603	70.0%	$37,212	74.0%
1R	4172.04	40.0%	$79,249	60.0%	$76,630	60.0%	$59,337	67.0%	$77,438	47.0%	$63,007	72.0%	$120,447	54.0%
2R	3407.91	48.0%	$55,864	49.0%	$59,275	54.0%	$40,125	42.0%	$59,880	46.0%	$43,705	44.0%	$103,022	48.0%
3R	1242.24	46.0%	$21,369	66.0%	$21,334	64.0%	$14,287	74.0%	$18,445	46.0%	$16,102	74.0%	$32,511	43.0%
4R	4354.45	41.0%	$85,267	45.0%	$88,342	36.0%	$65,512	50.0%	$99,917	46.0%	$70,485	49.0%	$164,652	48.0%
5R	2434.00	36.0%	$39,647	53.0%	$43,634	49.0%	$34,822	63.0%	$48,929	33.0%	$37,787	59.0%	$62,048	32.0%

Table A.12 Merchantable volume and sawlog volume estimates with coefficients of variation for all measured stands.

Estimated Stand Level Sawlog Volume and Variability														
	Merchantable Volume		Form and Risk**		Form and Risk*		Stick Cruising		Acceptable Sawlog Grade		Merchantable Stopper Heights		Forest Vegetation Simulator	
Stands	Volume	CV	Volume	CV	Volume	CV	Volume	CV	Volume	CV	Volume	CV	Volume	CV
1A	1541.51	44.0%	378.09	76.0%	438.80	72.0%	290.38	59.0%	574.51	56.0%	337.45	60.0%	734.69	64.0%
2A	1800.72	29.0%	684.91	71.0%	721.69	64.0%	486.01	96.0%	715.12	79.0%	503.47	104.0%	1162.31	36.0%
3A	1315.50	46.0%	451.41	84.0%	399.31	97.0%	303.12	96.0%	498.07	94.0%	336.87	96.0%	716.26	67.0%
4A	1948.42	54.0%	575.74	69.0%	646.73	71.0%	350.66	95.0%	502.00	98.0%	392.61	86.0%	1071.10	88.0%
5A	1343.85	53.0%	384.46	103.0%	353.12	116.0%	350.05	79.0%	511.12	70.0%	371.73	83.0%	718.61	78.0%
1R	4172.04	40.0%	1557.70	74.0%	1502.07	82.0%	1010.18	98.0%	1493.12	91.0%	1112.63	101.0%	2552.14	48.0%
2R	3407.91	48.0%	850.18	59.0%	925.17	67.0%	435.74	58.0%	928.81	58.0%	541.63	54.0%	2040.94	47.0%
3R	1242.24	46.0%	400.07	73.0%	373.70	80.0%	223.71	118.0%	382.54	111.0%	288.04	111.0%	781.72	40.0%
4R	4354.45	41.0%	1075.70	51.0%	1142.76	42.0%	705.72	68.0%	1490.56	66.0%	819.97	64.0%	2910.01	48.0%
5R	2434.00	36.0%	699.68	150.0%	783.15	132.0%	608.15	168.0%	919.41	121.0%	701.16	153.0%	1123.24	42.0%

Table A.13 Bioclimatic variable tested for significance from the climateNA data source.

MAT: mean annual temperature (°C)
MWMT: mean temperature of the warmest month (°C)
MCMT: mean temperature of the coldest month (°C)
TD: difference between MCMT and MWMT, as a measure of continentality (°C)
MAP: mean annual precipitation (mm)
MSP: mean summer (May to Sep) precipitation (mm)
AHM: annual heat moisture index, calculated as (MAT+10)/(MAP/1000)
SHM: summer heat moisture index, calculated as MWMT/(MSP/1000)
DD_0: degree-days below 0°C (chilling degree days)
DD5: degree-days above 5°C (growing degree days)
DD_18: degree-days below 18°C
DD18: degree-days above 18°C
NFFD: the number of frost-free days
bFFP: the julian date on which the frost-free period begins
eFFP: the julian date on which the frost-free period ends
FFP: frost-free period
PAS: precipitation as snow (mm)
EMT: extreme minimum temperature over 30 years
EXT: extreme maximum temperature over 30 years
Eref: Hargreave's reference evaporation
CMD: Hargreave's climatic moisture index
MAR: mean annual solar radiation (MJ m-2 d-1) (excludes areas south of US)
RH: mean annual relative humidity (%)
Tave_wt: winter (Dec to Feb) mean temperature (°C)
Tave_sm: summer (Jun to Aug) mean temperature (°C)
PPT_wt: winter (Dec to Feb) precipitation (mm)
PPT_sm: summer (Jun to Aug) precipitation (mm)

(Hamann et al., 2013; Wang, 2015)

B – Supplementary Equations

Smalian's Formula

$$V = hc(\frac{d_0^2}{2} + d_2^2 + \cdots + 2d_{n-1}^2 + \frac{d_n^2}{2})$$

Where: V = Volume (m^3)
d_i = diameter (cm) at ith measurement point
h = distance (m) between measurement points
c = diameter to cross-sectional area conversion factor (0.005454 for imperial units and 0.00007854 for SI units)

Kozak Taper Equation

$$\widehat{d}_i = \alpha_0 D^{\alpha_1} H^{\alpha_2} X_i^{\beta_i z^4 + \beta_2(1/e^{D/H}) + \beta_3 X^{0.1} + \beta_4(1/D) + \beta_5 H^Q + \beta_6 X}$$

Where:

$\widehat{d}_i$ = predicted inside or outside bark diameter at h_i height from ground (cm)
D = outside bark diameter at breast height (cm)
H = total tree height (m)
h_i = height from ground (m)
$z_i = h_i/H$, relative height from ground
$p = 1.3/H$ (relative breast height)
$Q = 1 - z^{1/3}$
$X = (1 - z^{1/3})/(1 - p^{1/3})$
$\alpha_0, \alpha_1, \alpha_2, \beta_1 \ldots \beta_6$ = species specific regression coefficients

Modified Height Imputation Model

$$H_{ijsk} = 4.5 + \exp(\alpha_a + b_i + b_j + b_s + \frac{Y}{1 + d_{ijsk}})$$

Where:

H_{ijsk} = total tree height (m) for the kth tree of s species on the ith plot on the jth site
b_s, b_i and b_j are random effects for the s species on th ith plot and jth site, respectively
α and Y are species specific coefficients
d = diameter at breast height in inches (cm)

International 1/4 Inch Log Rule

$$BF = .0497621 * (a * b^2) + 0.006220 * (a^2 * b) - 0.185476 * (a * b) + 0.000259 * a^3 - 0.011592 * a^2 + .04222 * a$$

Where:

BF = Board Feet

a = Log Length (ft)

b = Small End Log Diameter (in)

C- Field Protocols

All plots will be laid out using a systematic grid with 100m spacing using a random starting point prior to conducting the inventory. The approximate sampling intensity is 1 plot per forested hectare. Before conducting the inventory, the numbers 1:4 will be assigned to each plot point using a random number generator. The order that each methodology will be utilized will be determined by the assigned random numbers, with each number representing the corresponding methodology described below. A 20 BAF prism will be used for each stand. The definitions for what can be considered sawlog grade, pulp grade, or cull is given at the end of this document.

First measurements:

Plot center will first be recorded using a GPS device. A variable radius sampling method will be used at each plot. Beginning from true north at plot center and moving clockwise around plot center a 20 BAF prism will be used and all 'in trees' will have their species and diameter at breast height measured. Only live trees with a DBH of 10cm or greater will be recorded. A second sweep will be made from plot center with an 80 BAF prism and all 'in trees' will have their total height (top of crown) recorded. In order to capture the actual canopy height, the actual height of the crown in feet will be recorded regardless of the stems angle relative to the ground. If a tree forks below breast height each stem after the fork will be considered as a unique tree, but the low fork form class being noted for each stem. The walk-through method will be used for any plots near the stand boundary (Iles, 2003).

Each method below will be timed using a stopwatch beginning from the location of true north until the final 'in tree' at each plot is recorded. The time it takes to complete each method will be recorded at every plot.

1. ASG/USG:

Acceptable Sawlog Grade (ASG) will be defined as any tree with a high-quality stem and high vigor or a high-quality stem of low vigor. In order to be defined as ASG the stem must contain at least one sawlog that is 2.45m or longer in length. The existence of a sawlog meeting the minimum length will be visually estimated.

This methodology is an adaptation of the simplified classification system as described in the Ontario Tree Marking Guide (2004). This classification scheme recognizes that stem quality is a stronger predictor of product recovery; as while quality and vigor are related, vigor primarily reflects the likelihood of future growth and mortality. While vigor clearly is a required consideration for silvicultural decisions, our goal is the assessment of the standing value of the wood at present day in stands that are to be harvested within six months.

No estimate of the exact length or volume of sawlog recovery will be made for ASG trees. The trees will simply be recorded as sawlog quality stems and their merchantable sawlog volume will be calculated as the total volume from the stump to the minimum merchantable sawlog diameter of the stem, which is based on its species.

All trees that cannot be considered ASG will be marked as unacceptable sawlog grade and labeled as either pulp or cull quality.

2. Stick Cruse:

For each tree, a product call will be made for every 2.45m section of the stem beginning at a .5m stump and ending at the merchantable length of the stem where it reaches a 10cm top diameter. The product classes for each section can either be “Sawlog”, “Pulp”, or “Cull”. The length of

each eight-foot section, the height at which the minimum merchantable diameter of the tree is reached, and the corresponding product calls will be visually estimated. The total number of sawlogs predicted for each stem will be recorded for each ‘in tree’.

3. Merchantable Stopper Height:

Starting at true north and moving clockwise from plot center, the length of the sawlog quality portion of each stem will be measured using a Haglof clinometer. The height at which sawlog potential begins will be recorded. This height will be recorded as the ‘stump’ height. For all trees where a 2.45m sawlog can be recovered at any point along the stem, a second height will be recorded indicating the length of the sawlog portion of the stem. If more than one 2.5m sawlog can be recovered, but there is a small section between them that will be bucked due to defect, the entire length of the section will be assumed to be of sawlog quality. Trees that can produce no sawlog or pulp material will be considered cull (Figure X).

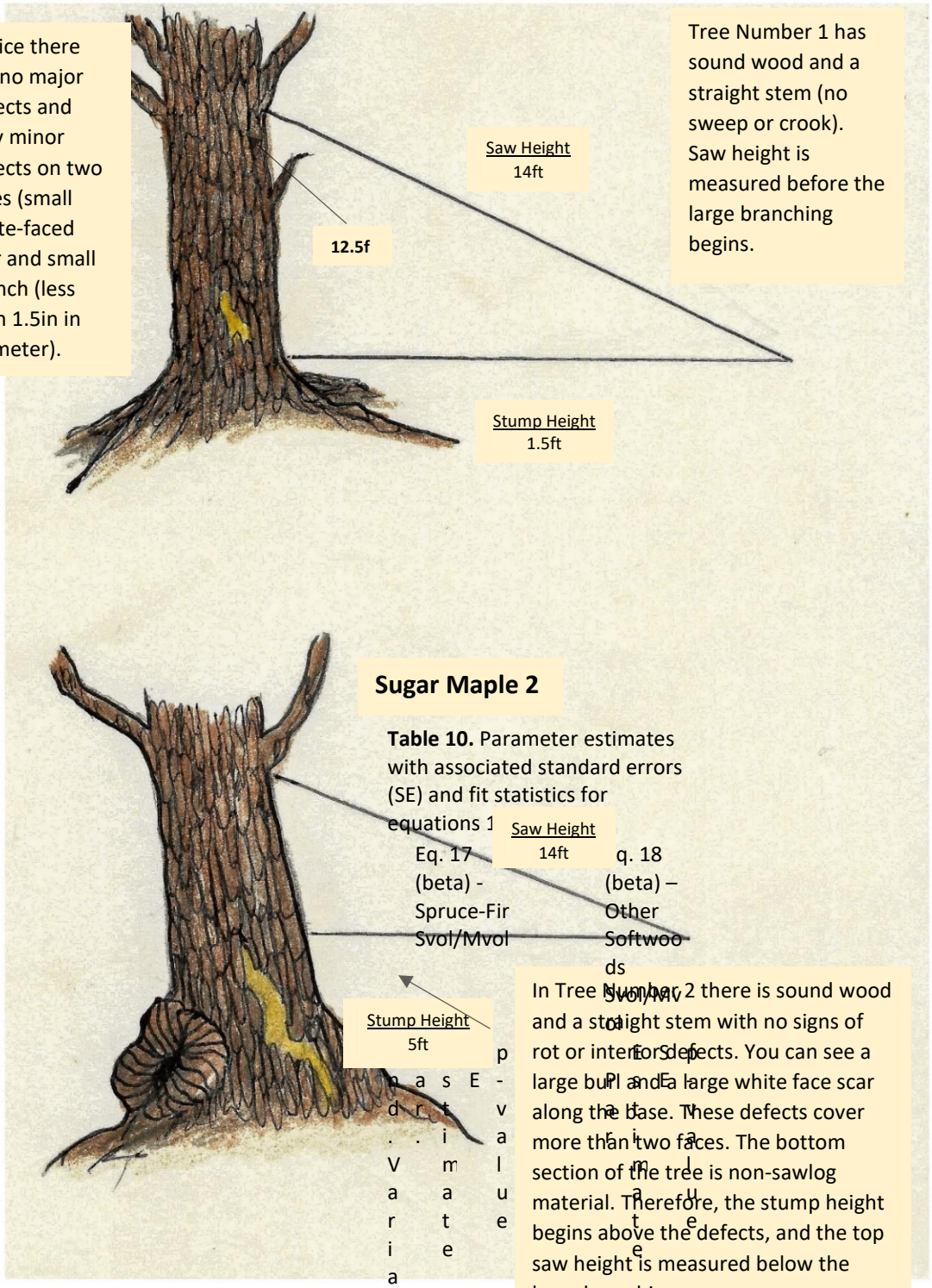


Figure A.1 Demonstration of merchantable stopper height measurements.

Table 10. Parameter estimates with associated standard errors (SE) and fit statistics for equations 1

Parameter	Eq. 17 (beta) - Spruce-Fir Svol/Mvol	Eq. 18 (beta) – Other Softwoods Svol/Mvol
Estimate	2.760021	25.002011
SE	0.701	1.120
p-value	<0.0001	<0.0001
Adj. R^2	0.7252	0.8180

4. Form/Risk:

Starting at true north and moving clockwise, the Northern Hardwoods Research Institute form and risk class of each tree will be recorded. The form and risk classes will be those defined in table 1.1.

Sawlog Definition

Portions of the stem that are considered potential sawlog material are any that maintain at least 2 clear faces over a 2.45m portion of the stem, and where there is not a 'disqualifying' defect.

Clear Faces

Clear faces will be defined as faces without:

- without significant visible bulging
- no branches larger than 1in in diameter
- no burls
- no healed white face scars
- no twisting seams
- no cavities
- no cankers
- no crooks or sweep that is visually estimated to be more than 12cm over 2.45m.
- no scars that are over ¼ the length of the log.

Disqualifying defects:

Disqualifying defects include a broken or dead limb larger than 5cm in diameter along the potential sawlog portion of the stem, visible rot, significant cavities, or excessive sweep or crook at any place along the potential sawlog portion of the stem. Any tree that displays clear signs of internal defect or rot, large cavities along the stem, long twisted seams, long unhealed or

blackface scars, rot along the stump of the tree, visible fungal growth or disease will be considered to not have any sawlog potential.

Pulp

Pulp sections will be any that do not have clear visible rot or disease and where the sweep is not so significant that it clearly could not fit through an 80cm diameter culvert.

Cull

All portions of the tree that are not sawlog or pulp grade will be considered cull.

BIOGRAPHY

Ryan Smith is from Poughkeepsie, NY. He graduated from Clark University in Worcester, MA with a B.A. in International Development and Social Change in 2009, and an M.A. in Community Development and Planning in 2012.

Ryan worked for several years as a union organizer in New York, New York with the Hotel and Motel Trades Council. Ryan is a proud nine-year journeyman member of the International Association of Bridge, Structural, Ornamental, and Reinforcing Ironworkers' Union Local 40 based in New York City.

When not pursing his interests in forestry, Ryan enjoys running, fishing, rock climbing, following the NY Jets, and spending time with his wife, family, and friends.

He is a candidate for the M.S. degree in Forest Resources from the University of Maine in December 2022.